

Computer Vision 1

Shape from X & 3D Reconstruction

Dr. Martin R. Oswald

Lecture overview

Week		Date	Presenter	Lectures	Excerise / Labs
1	36	02/09/2024	Arnoud Visser	1. Introduction and Course Logistics; Camera model	
1	36	05/09/2024	Arnoud Visser	2. Sensor and image formation Color	
2	37	09/09/2024	Arnoud Visser	3. In place image processing Photometric stereo	
2	37	12/09/2024	Arnoud Visser	4. Convolution and image filtering	
3	38	16/09/2024	Arnoud Visser	5. Feature: edge and corner	
3	38	19/09/2024	Arnoud Visser	6. Motion and optical flow	
4	39	23/09/2024	Martin Oswald	7. Global transform, Image stitching, alignment	
4	39	26/09/2024	Martin Oswald	8. Basics of Recognition 1	Online Lectures
5	40	30/09/2024	Martin Oswald	9. Basics of Recognition 2	
5	40	03/10/2024	Martin Oswald	10. Convolutional neural networks (Basic)	
6	41	07/10/2024	Martin Oswald	11. Convolutional neural network (Training and testing)	
6	41	10/10/2024	Martin Oswald	12. Stereo, 3D reconstruction	
7	42	14/10/2024	TBA	13. Guest lectures: Sezer Karaoglu, Gertjan Burghouts	
7	42	17/10/2024	TBA; Martin & Arnoud	14. Guest lecture: D. Tzionas; Course Wrap-up, Exam Q&A	
8	43	21/10/2024	Your turn!	Final exam	

Today



dr. Sezer Karaoglu
3DUniversum
Co-Founder & CTO

Monday



dr. Gertjan Burghouts
TNO
researcher in deep learning &
computer vision

Monday



dr. Dimitrios Tzionas
University of Amsterdam
Assistant Professor

Thursday

Very easy questions from each of the speakers talk will be in the exam, each lecture worth 1 pt

Today's Agenda



- Shape from X
- 3D Reconstruction
 - 2-view case: Stereo reconstruction
 - Camera calibration
 - Epipolar geometry
- Structure from Motion

- Chapter 12, 13, 14
- 3D Vision is mostly covered in CV2, a brief introduction in CV1

Recap

Faster R-CNN: Learnable Region Proposals

Faster R-CNN is a
Two-stage object detector

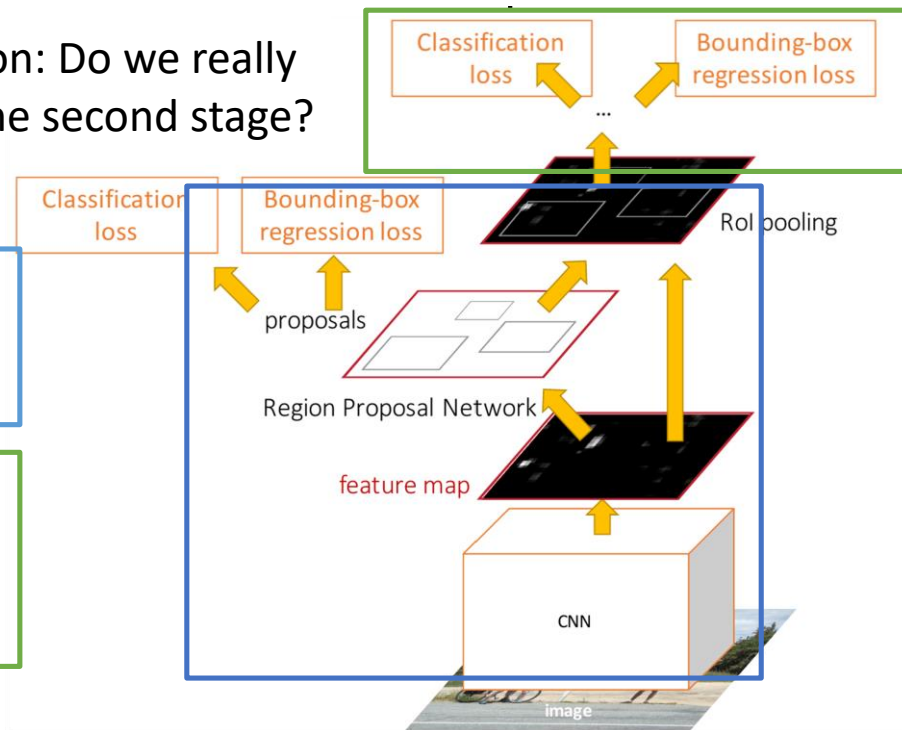
First stage: Run once per image

- Backbone network
- Region proposal network

Second stage: Run once per region

- Crop features: RoI pool / align
- Predict object class
- Prediction bbox offset

Question: Do we really
need the second stage?

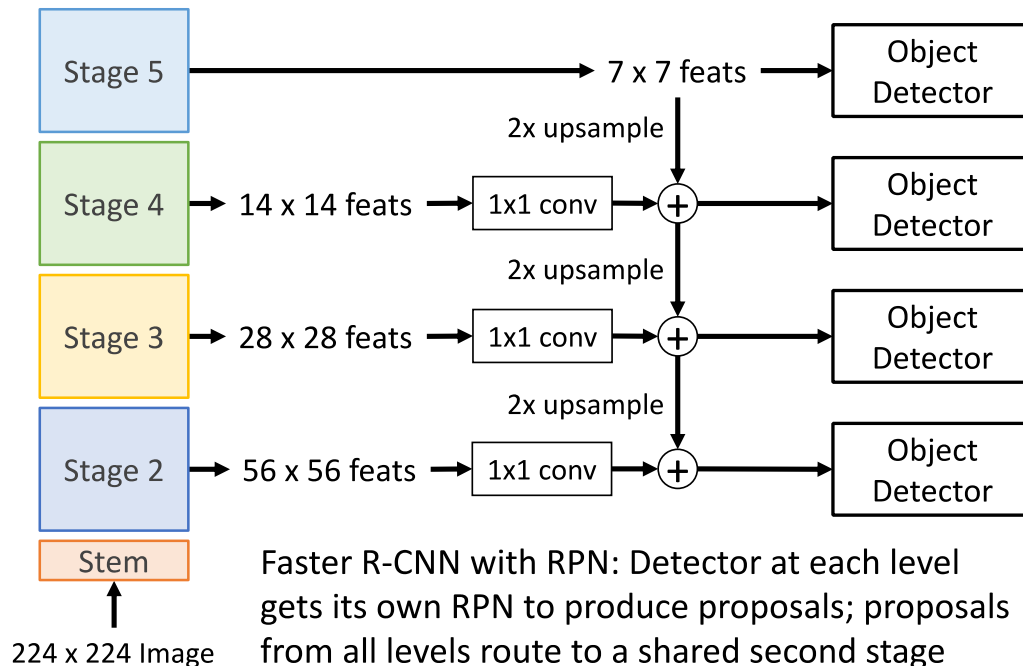


Scale Variations: Feature Pyramid Network

Add *top down* connections that feed information from high level features back down to lower level features

Efficient multiscale features where all levels benefit from the whole backbone! Widely used in practice

Lin et al, "Feature Pyramid Networks for Object Detection", ICCV 2017



Single-Shot Detection

Anchor based single-shot detectors work with following steps:

- A. Divide image in grid cells
- B. Pre-define anchor-boxes
- C. Refine box coordinates
- D. Train class confidence
- E. Filter best predictions



Single-Shot Detection

E. Filter best predictions

Non-maximum suppression filters out predictions with low class confidence. Filter out boxes with confidence below threshold.

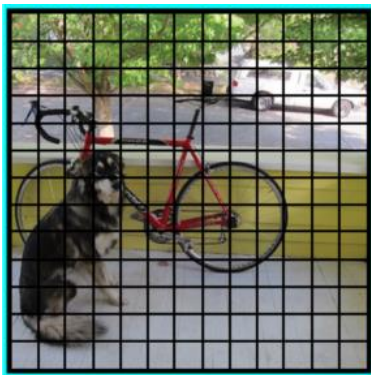
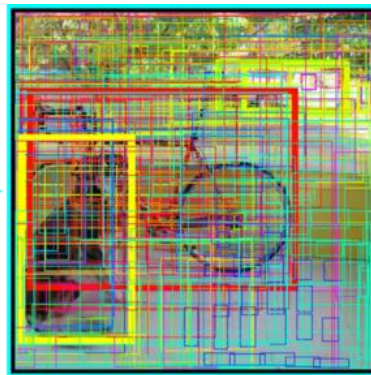
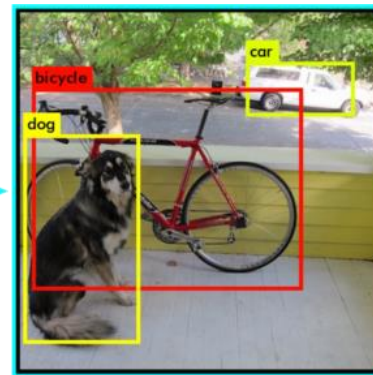


Image grid
 $S \times S$
 $S = 13$



Prediction
Every grid cell B boxes
 $B = 5$

So:
 $13 \times 13 \times 5 = 845$ boxes

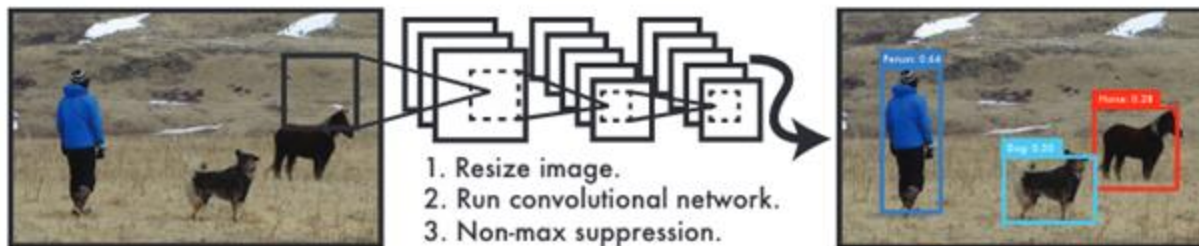


Thresholding:
Box confidence
score > 0.3
Non-Maximum suppression
Class predictions with highest
class confidence score

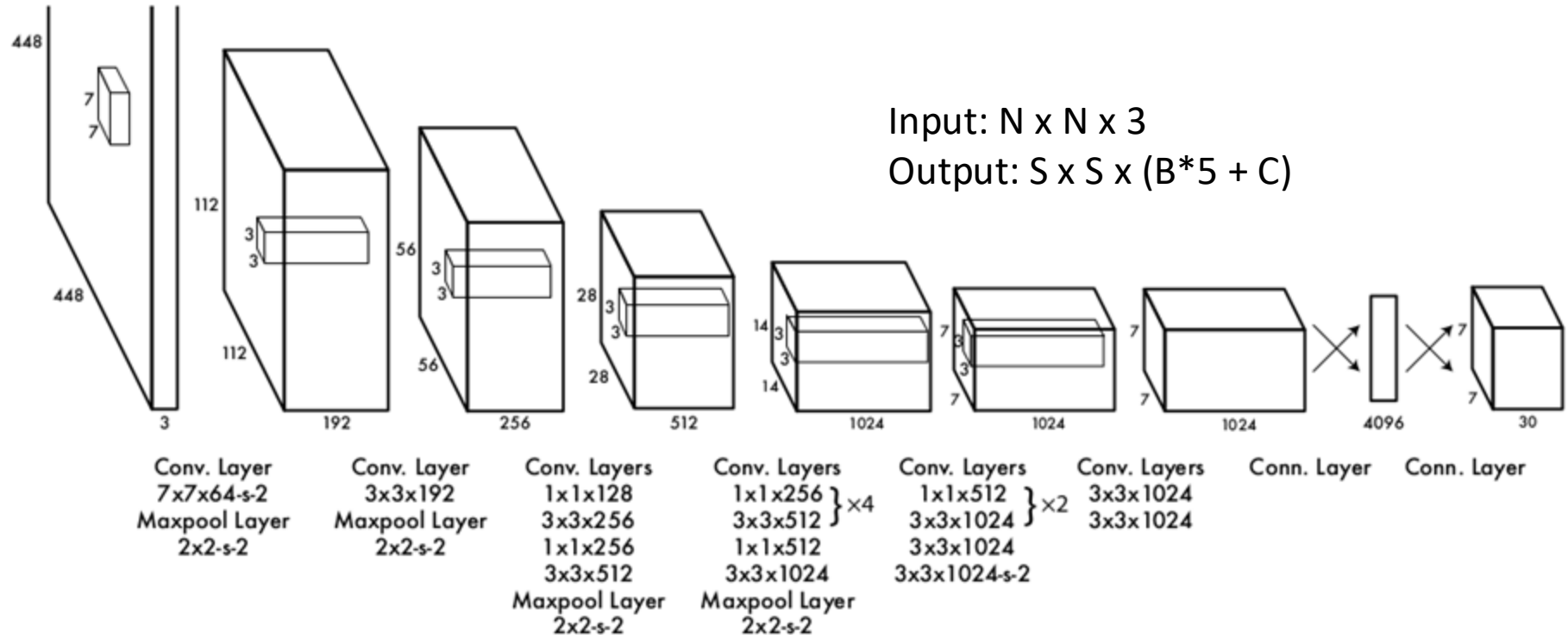
You Only Look Once (YOLO)



- Simple architecture
- Single regression problem, from pixels to bounding box coordinates and class probabilities:
simpler to optimize
- **FAST:** 45 FPS for YOLO and 150 FPS for YOLO-Fast
- Sees the global context, not just patches



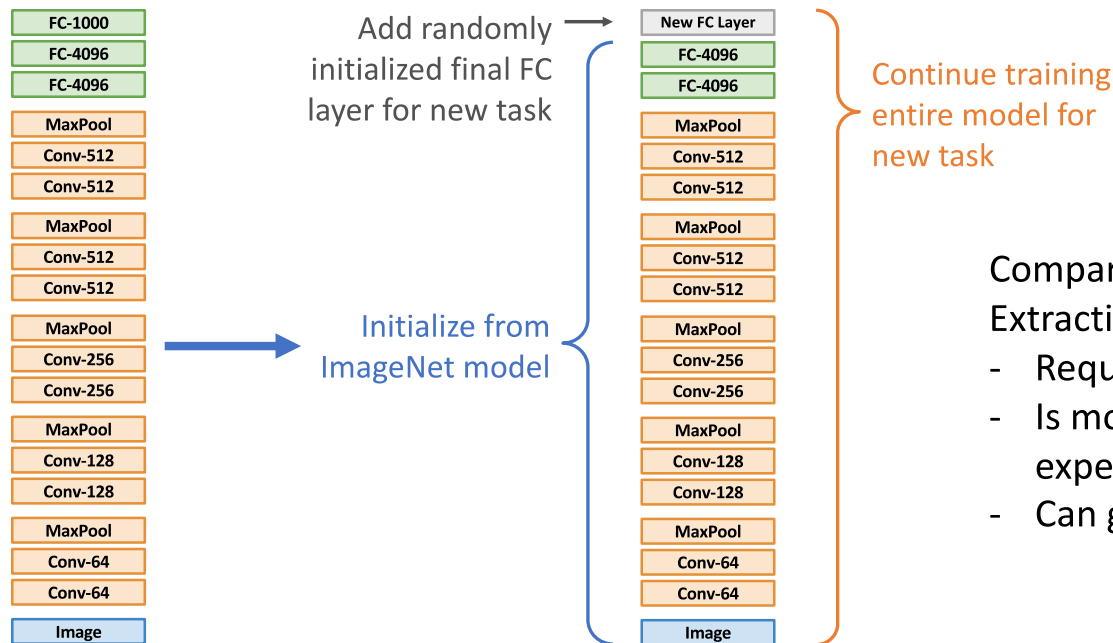
YOLO - Architecture



Transfer Learning

Transfer Learning: Fine-Tuning

1. Train on ImageNet

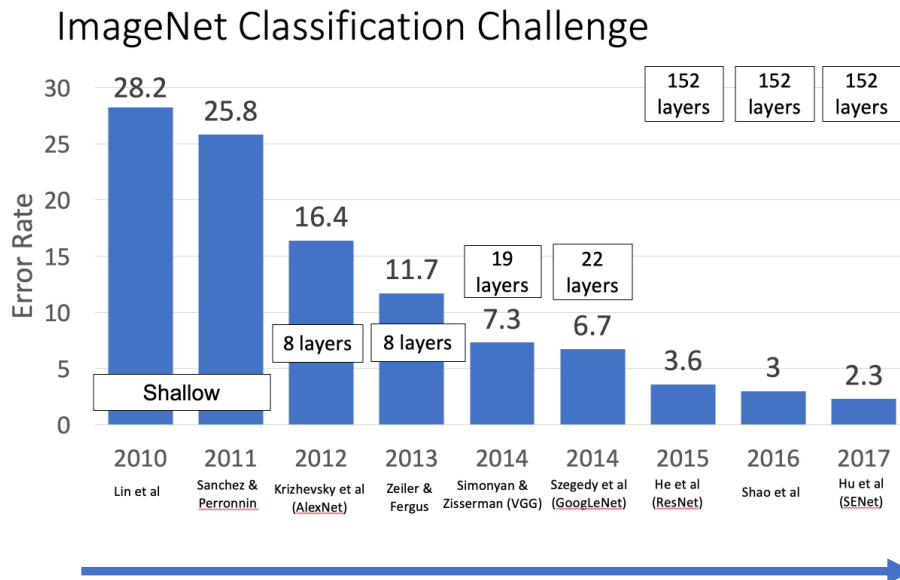


Compared with Feature Extraction, Fine-Tuning:

- Requires more data
- Is more computationally expensive
- Can give higher accuracies

Transfer Learning

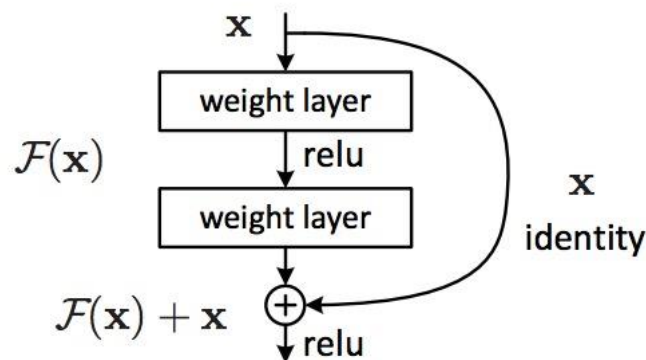
Transfer Learning: Architecture Matters!



Improvements in CNN architectures lead to improvements in many downstream tasks thanks to transfer learning!

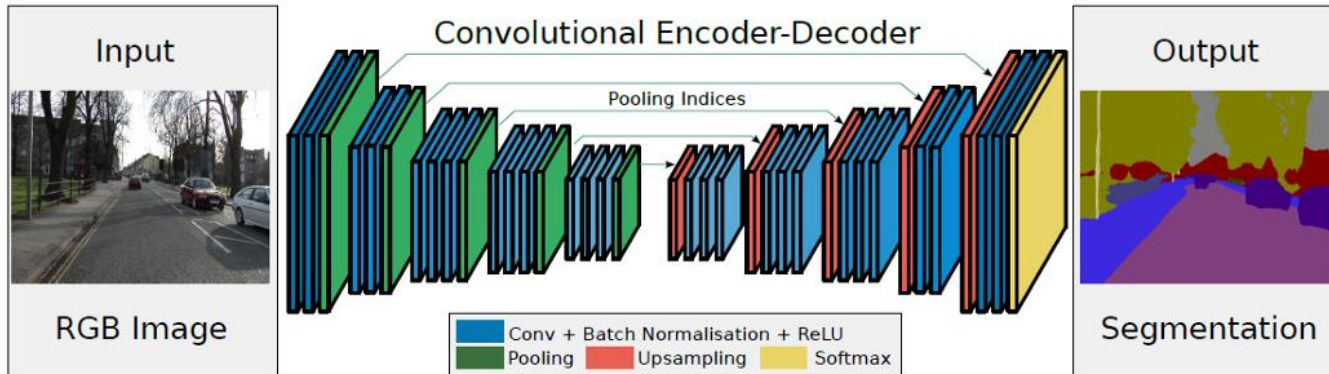
ResNet (ILSVRC 2015 winner)

- The residual module
 - Introduce *skip* or *shortcut* connections (existing before in various forms in literature)
 - Make it easy for network layers to represent the identity mapping

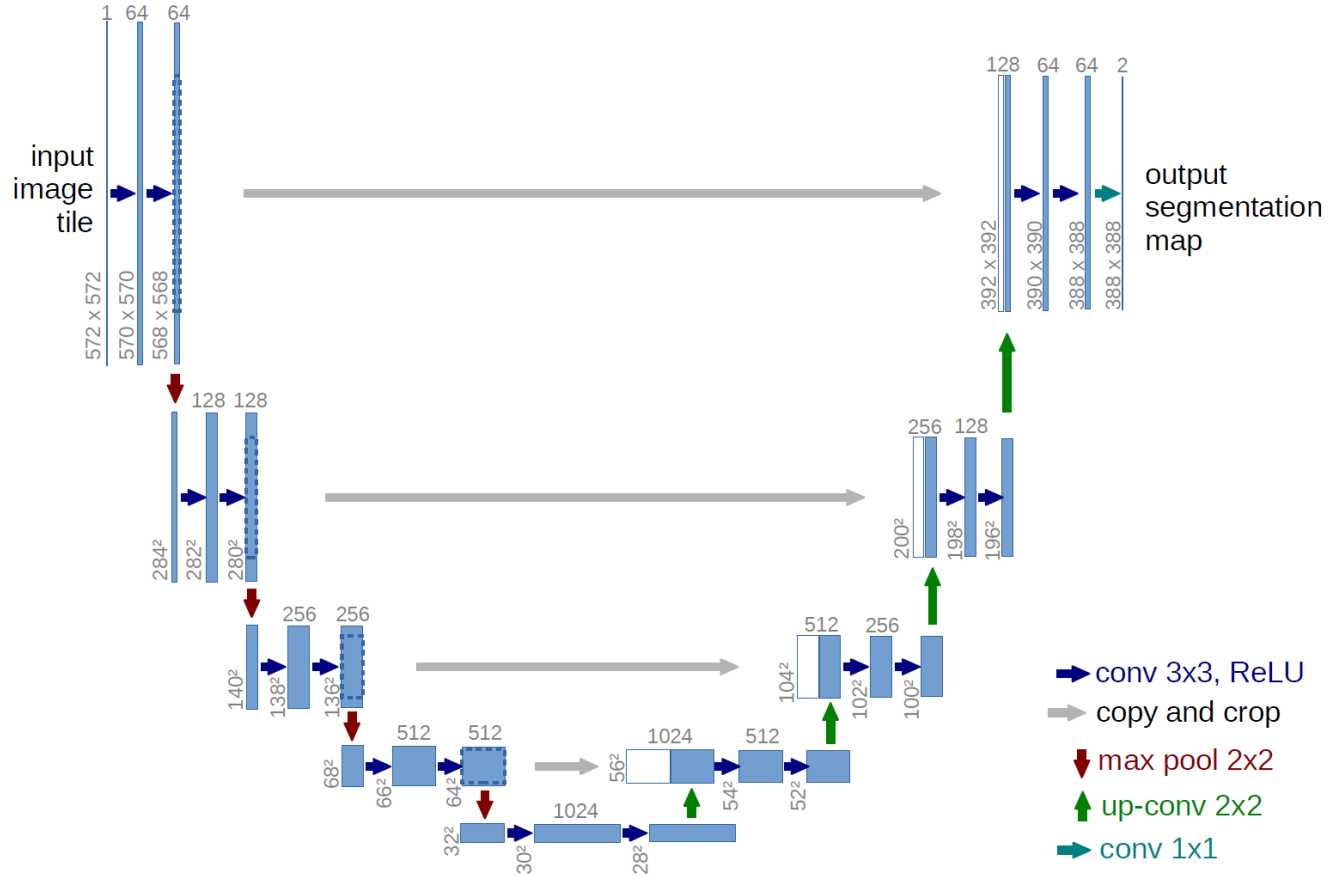


Encoder-Decoder Architecture - SegNet

Segnet uses an encoder-decoder architecture for semantic segmentation.

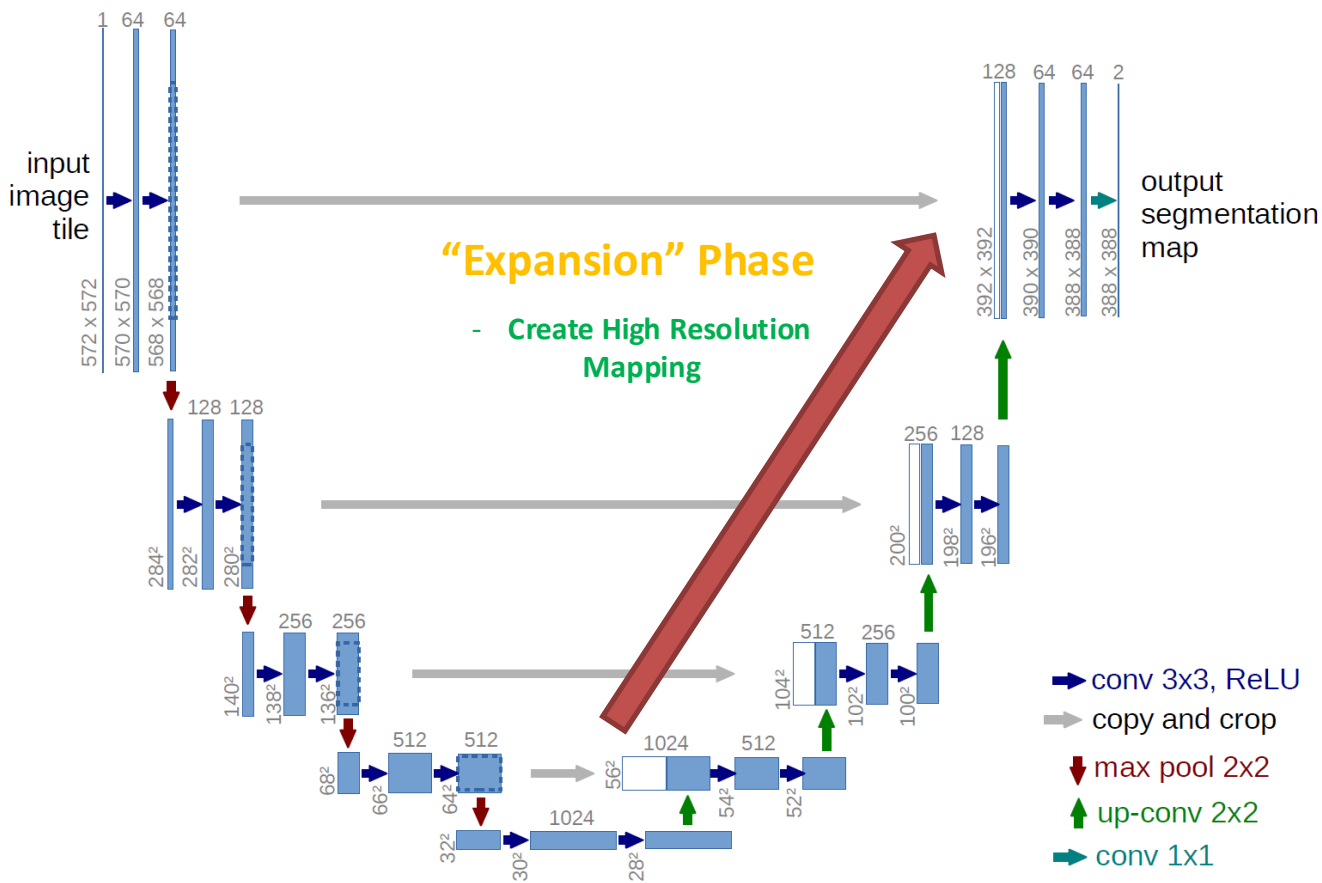


U-Net Architecture

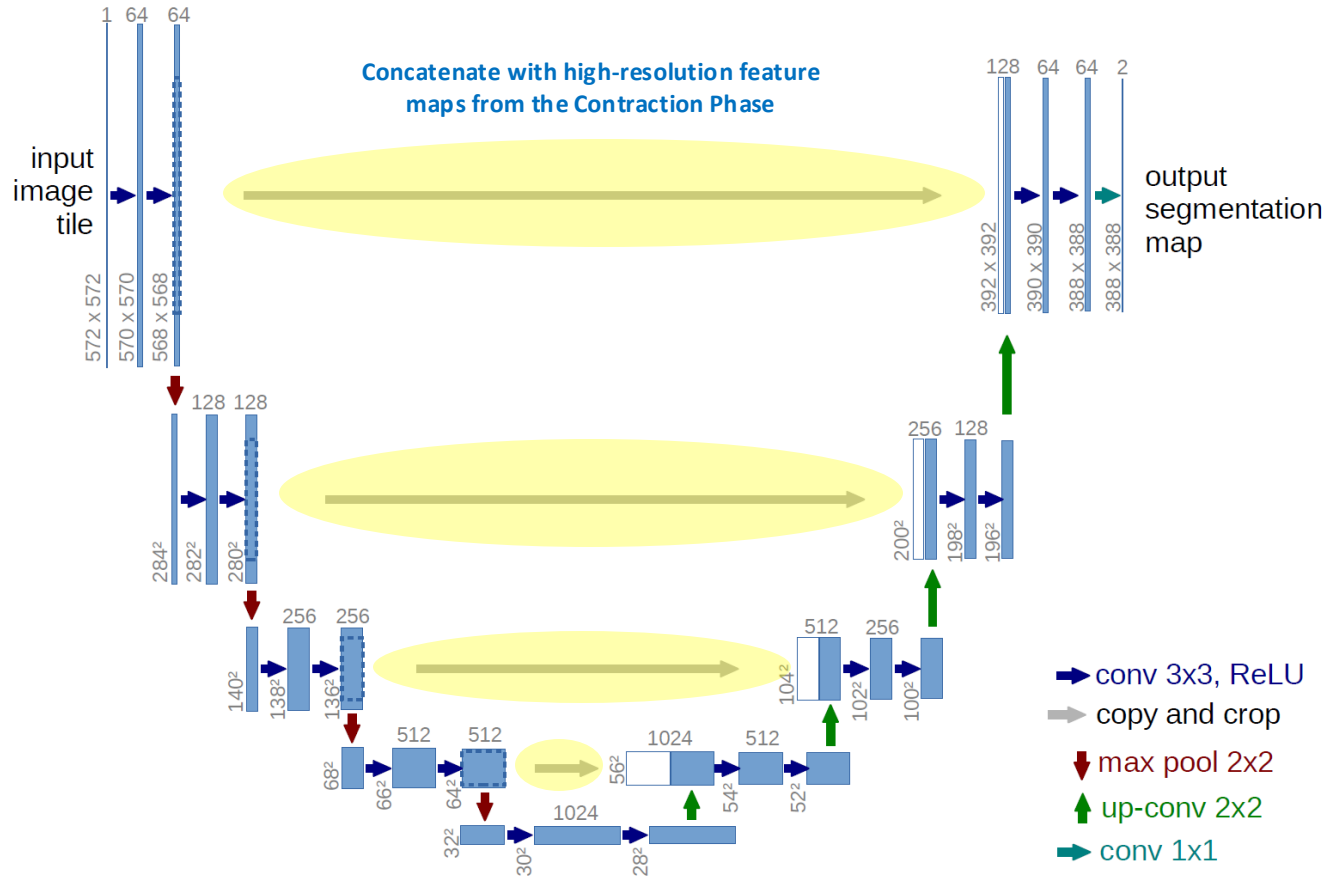




U-Net Architecture

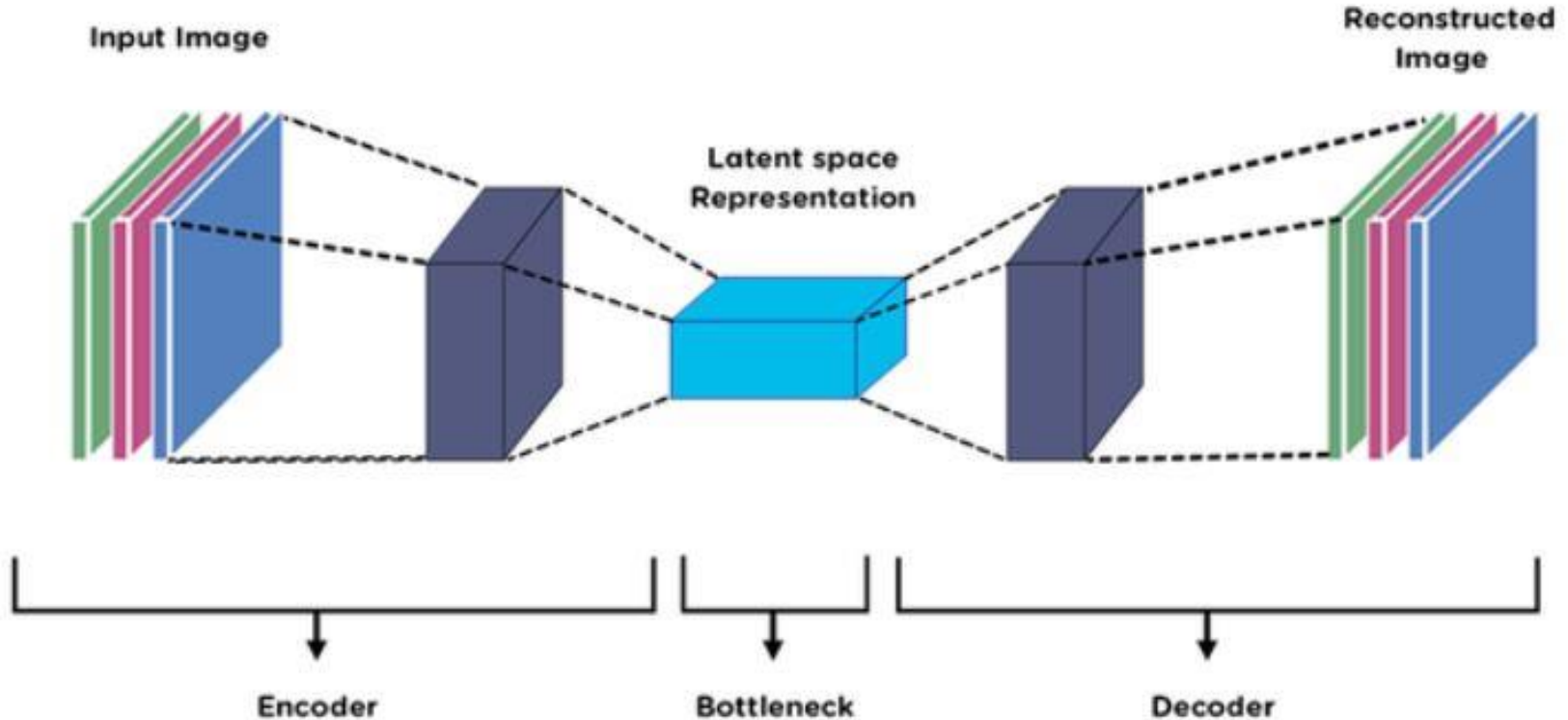


U-Net Architecture



Autoencoders

Latent space is of lower dimension than input.
So reconstruction typical not perfect.



Today's Agenda



- Shape from X
- 3D Reconstruction
 - 2-view case: Stereo reconstruction
 - Camera calibration
 - Epipolar geometry
- Structure from Motion

Motivation: Image Understanding

AN IMAGE IS WORTH A THOUSAND WORDS



Motivation: Scene Understanding

So far mostly 2D Computer Vision:

- What is in the image?
- Object categories
- Object counts
- 2D object locations (bbox/mask)

3D Computer Vision:

- Where are objects in the scene?
- Distances
- Relative 3D positions
- Spatial object relations
- Object sizes
- Geometric shape of environment

AN IMAGE IS WORTH A THOUSAND WORDS



Challenges in 3D Reconstruction



Julian Beever with his anamorphic Coke bottle

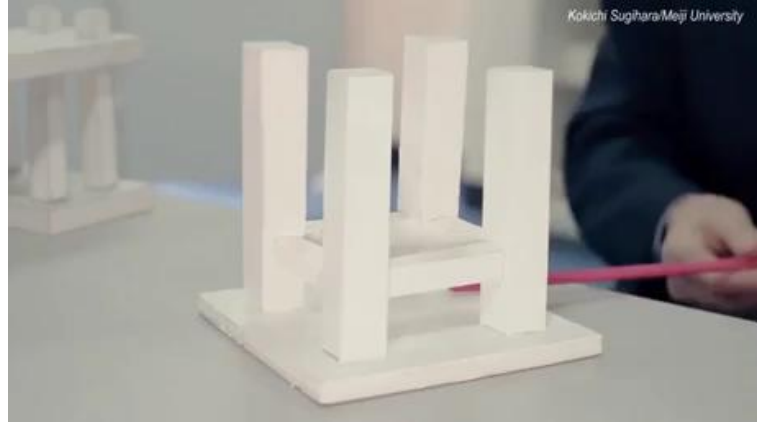


- Output of monocular depth estimator
- or: How to stop an autonomous car?

Challenges: View Ambiguity



Challenges: View Ambiguity



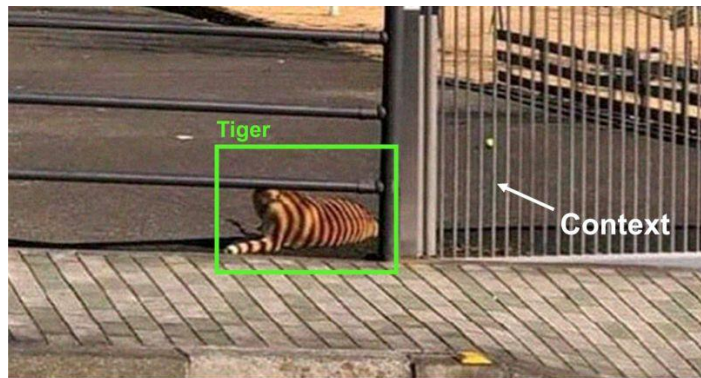
Challenges: Texture vs. Shape



Challenges in Localization



Challenges in Scene Understanding



Shape-from-X

X =

- Shading
- Texture
- Focus / Defocus
- Specularities
- Shadows
- Silhouettes
- Motion (/ Structure-from-Motion)
- Multiple Light Sources (photometric stereo)
- ...

Shape-from-Shading

- Goal is to reconstruct a 3D profile from a single image
- Typical assumptions:
 - Diffuse material with constant albedo
 - Known light source at infinity
 - Known camera at infinity (orthography)
- Exploits relationship between shading and surface normal direction

Challenges / Problems:

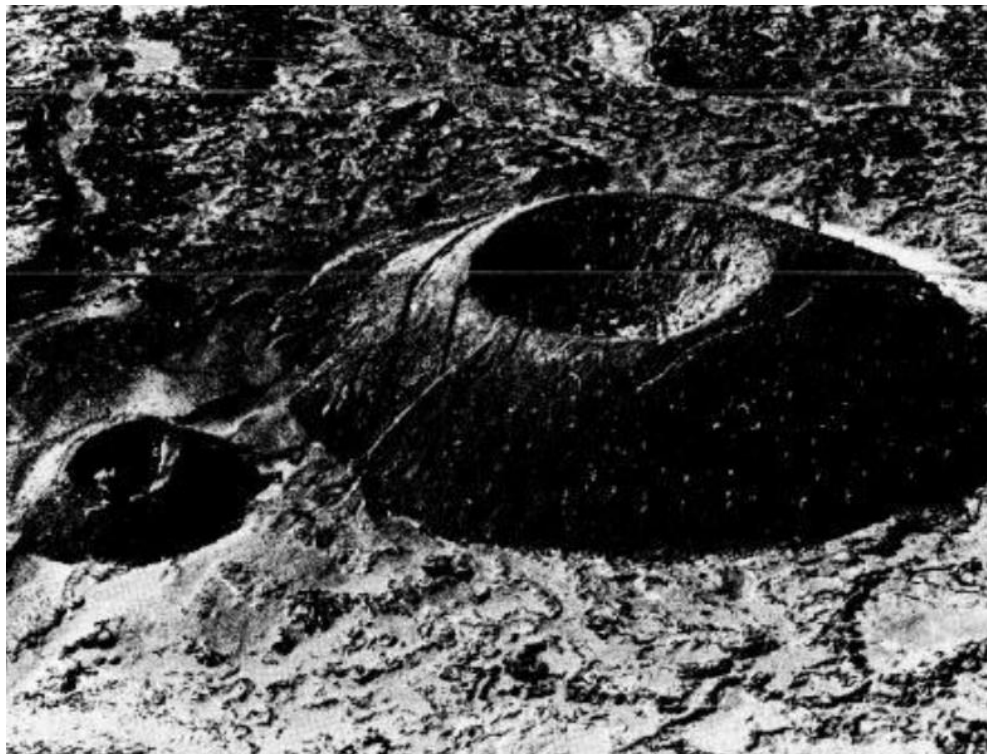
- Light direction ambiguities
- Non-Lambertian surfaces
- Texture vs. Shading differentiation
 - > works best with textureless, homogeneously colored objects



Image of the surface of planet Mars. Source: NASA

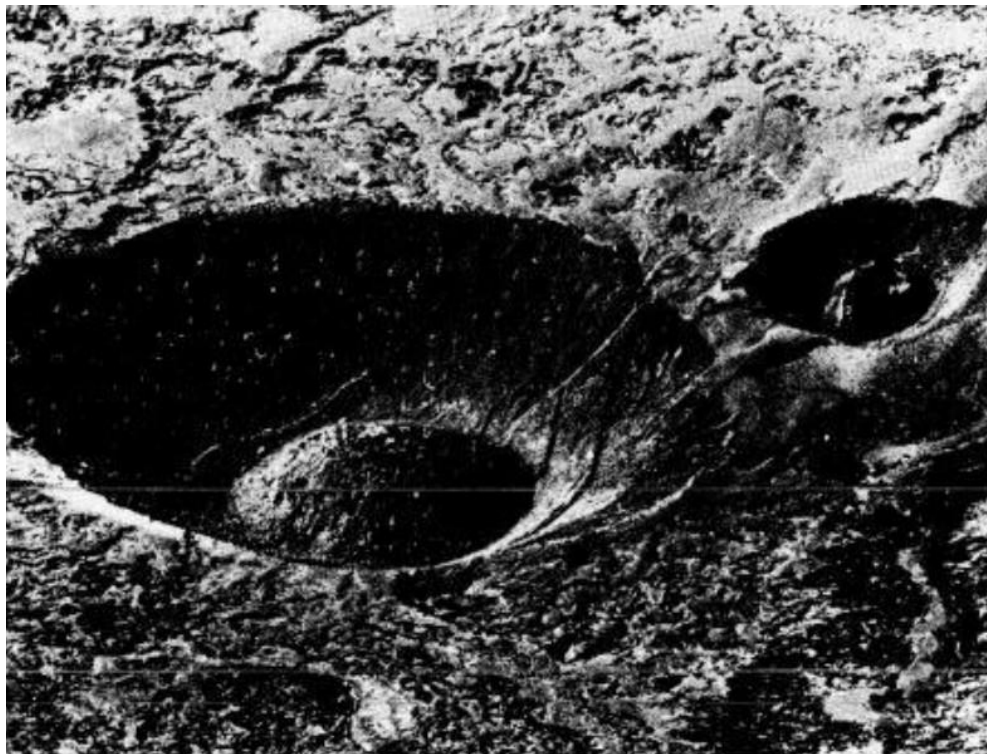
Shape-from-Shading: Ambiguities

Human vision resolves light direction ambiguities by assuming light is coming from above.

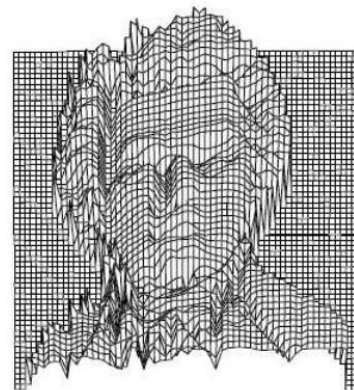
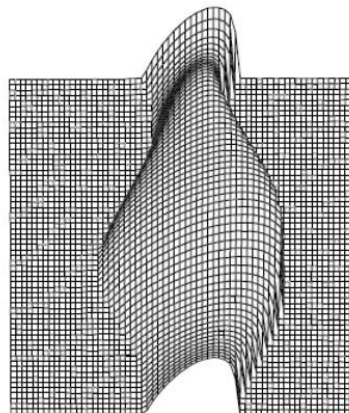
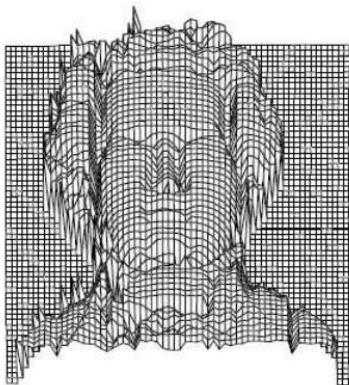
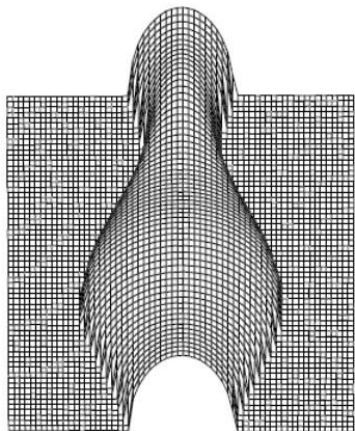
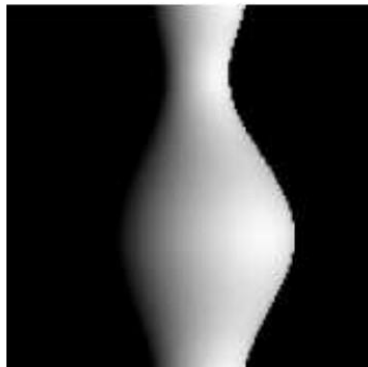
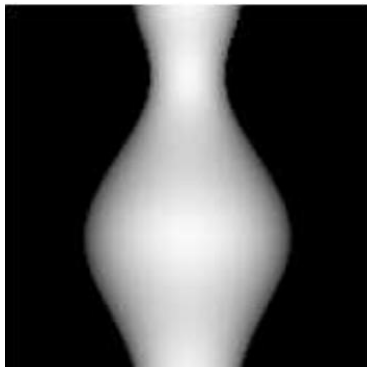


Shape-from-Shading: Ambiguities

Human vision resolves light direction ambiguities by assuming light is coming from above.



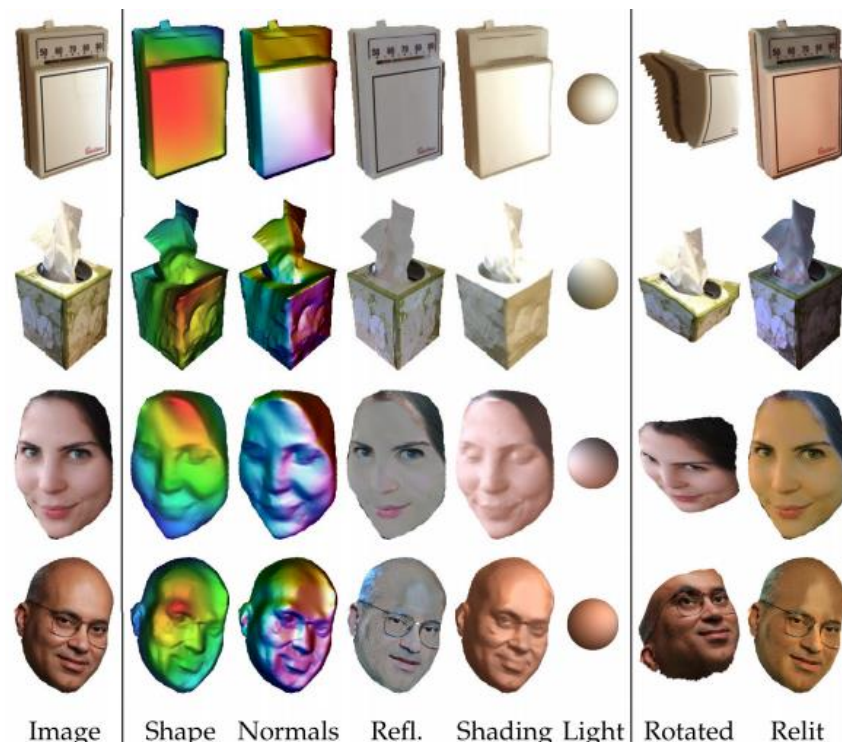
Shape-from-Shading – Early Results



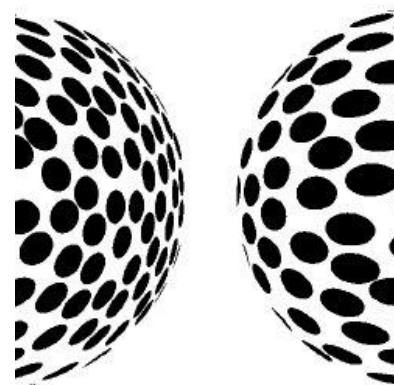
Shape-from-Shading – Results

SIRFS: Shape, Illumination, and Reflectance from Shading

- Modern version of SfS
- Input: Single masked RGB image
- Estimates depth, normals, reflectance, shading, lighting
- Does not assume known point light
- Does not assume constant color
- Much harder problem than SfS

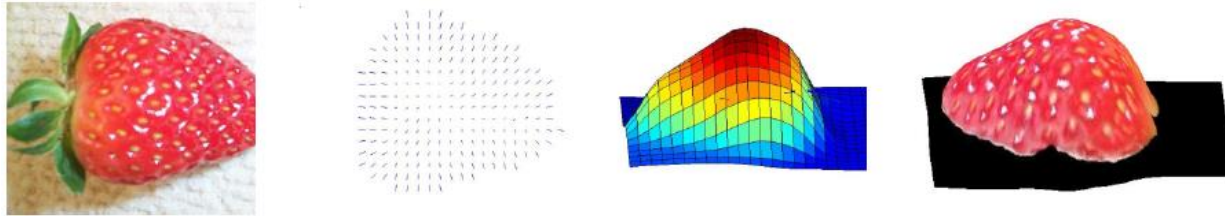


-



Shape-from-Texture

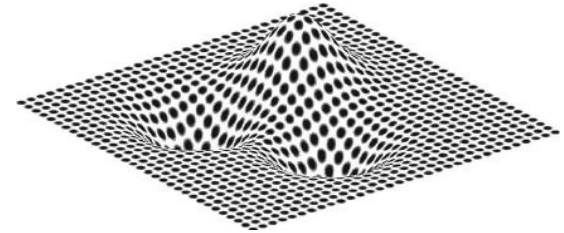
- Goal is to reconstruct a 3D profile from a single image assuming a repetitive texture pattern



the maximum likelihood estimate. It seems most likely that the discrepancies result from a local "basin of attraction" for learning that does not lead to a local maximum of the likelihood. The likelihood can be eliminated for the model with a correlation function in either of two ways. One way is to reduce the learning parameters (in the wake phase), while leaving the learning rate unchanged. As discussed in section 3.2, there is a good reason to think that this method will lead to the maximum likelihood, since the recognition model will then have time to learn the generative model perfectly. When the generative learning is reduced by setting $\alpha = 0.0005$ and $\alpha = 0.9995$, the maximum likelihood is indeed found in eight of eight test runs. The second way is to impose a constraint that prevents the maximum likelihood



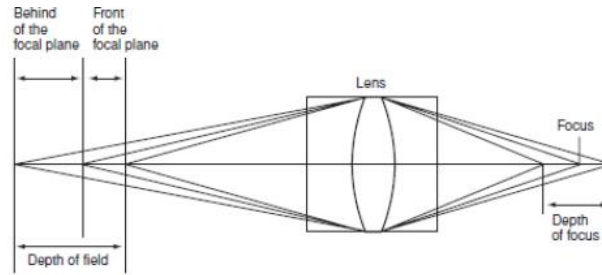
the maximum likelihood estimate. It seems most likely that the discrepancies result from a local "basin of attraction" for learning that does not lead to a local maximum of the likelihood. The likelihood can be eliminated for the model with a correlation function in either of two ways. One way is to reduce the learning parameters (in the wake phase), while leaving the learning rate unchanged. As discussed in section 3.2, there is a good reason to think that this method will lead to the maximum likelihood, since the recognition model will then have time to learn the generative model perfectly. When the generative learning is reduced by setting $\alpha = 0.0005$ and $\alpha = 0.9995$, the maximum likelihood is indeed found in eight of eight test runs. The second way is to impose a constraint that prevents the maximum likelihood



[Source: Gumsey et al., Journal of Vision, 2006]

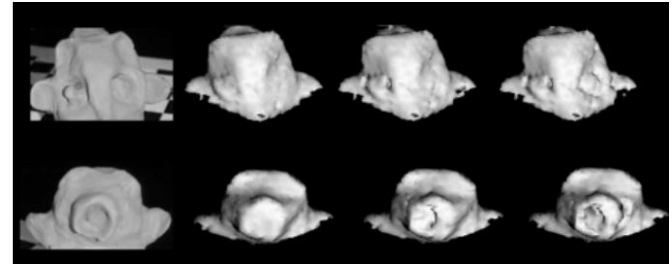
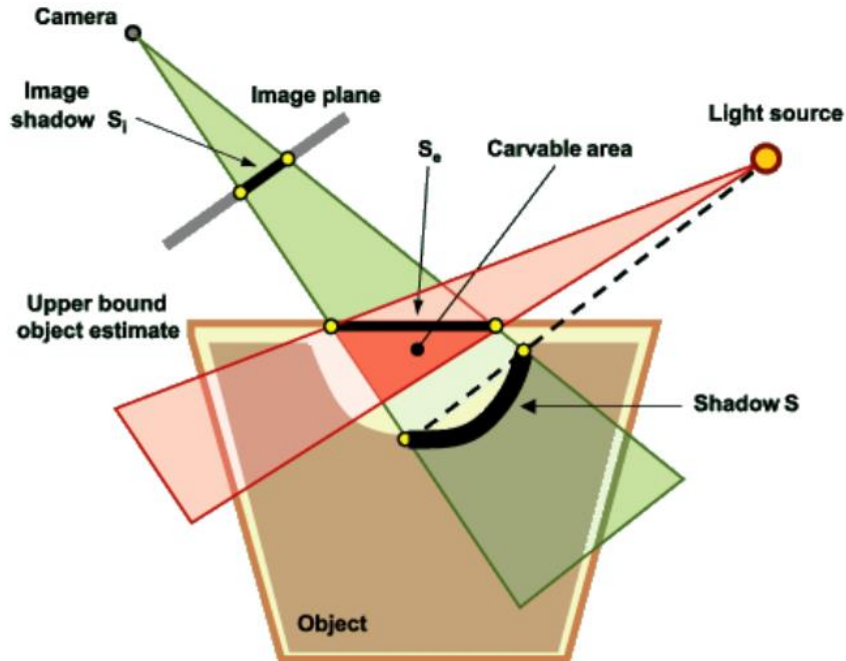
Shape-from-Focus

- Sweep through focus settings “most sharp” pixels correlate with the depth (most high frequencies)

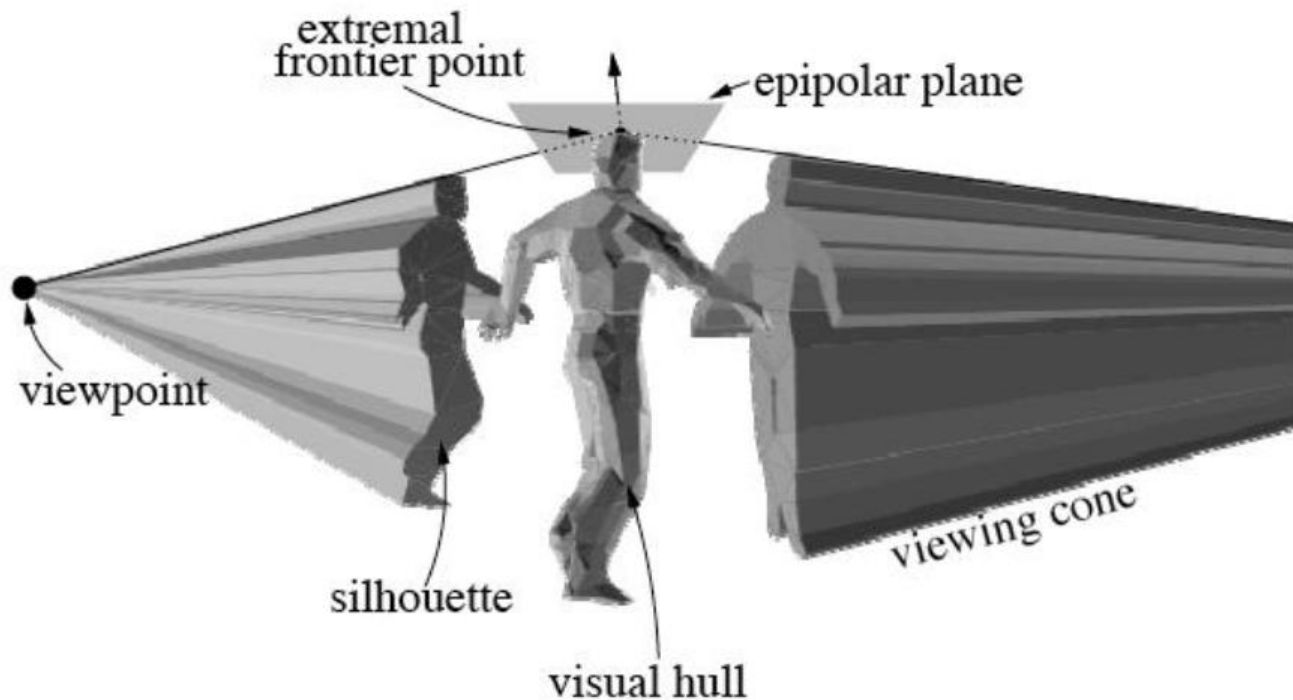


Shape-from-Shadows

- Detect shadows and use them to recover shape cues



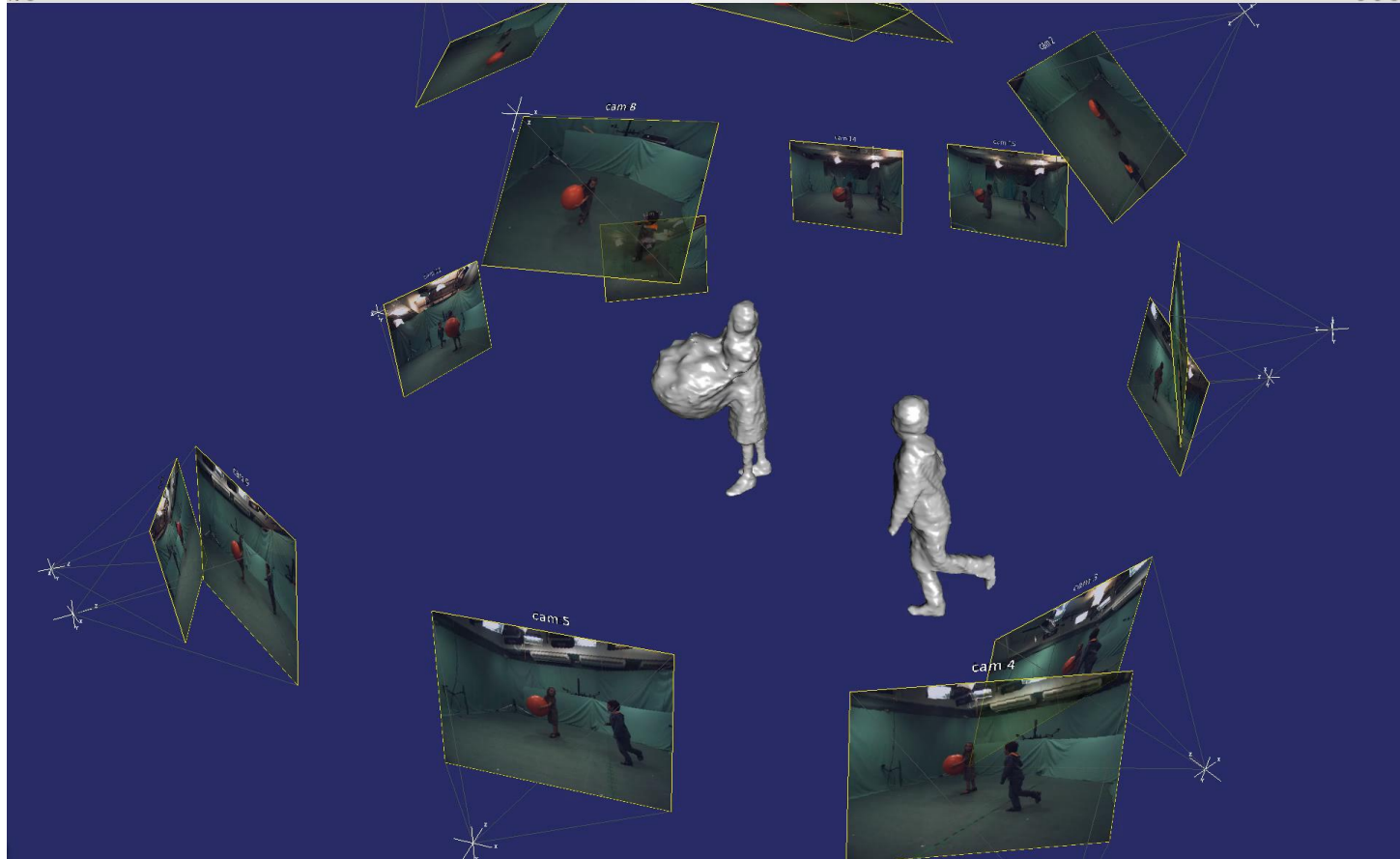
Shape-from-Silhouettes



Visual Hull: Intersection of all silhouette-based viewing cones
(with known cameras)

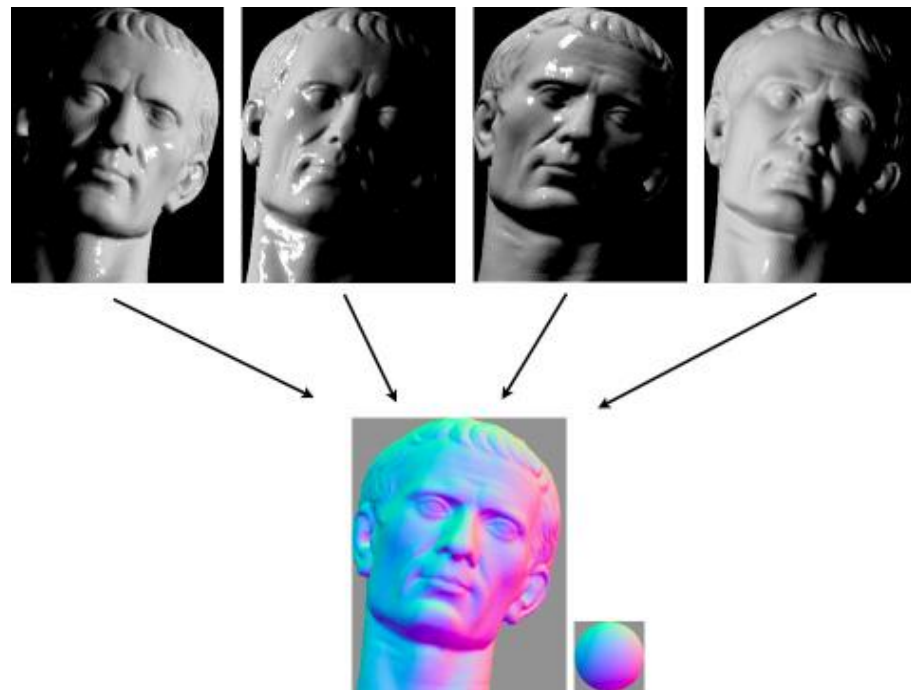


Shape-from-Silhouettes



Photometric Stereo

- Instead of smoothness constraints, add **more observations per pixel**
- Take **K images** of the object from **same viewpoint** (e.g., with a tripod) but with different (known) point light source each
- Per-pixel estimation of **normal** and **albedo** or **material**
- Also assumes far camera/light



Photometric Stereo – via Light Stage

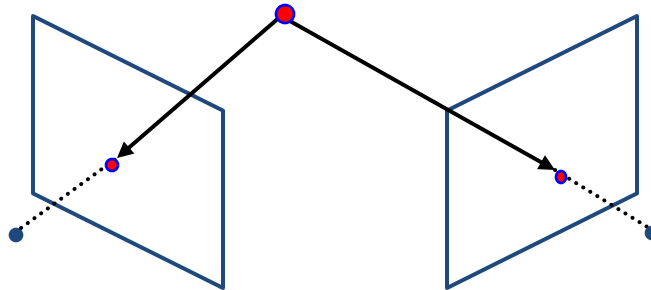


Today's Agenda



- Shape from X
- 3D Reconstruction
 - 2-view case: Stereo reconstruction
 - Camera calibration
 - Epipolar geometry
- Structure from Motion

Stereo Vision

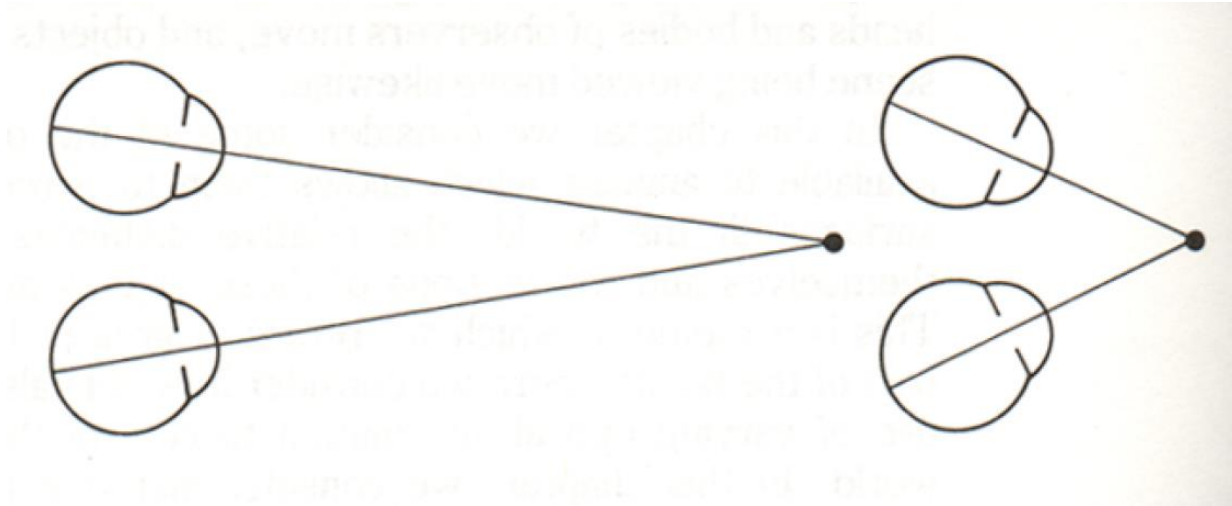




If stereo were critical for depth perception, navigation, recognition, etc.,
then rabbits would never have evolved.

Human stereopsis

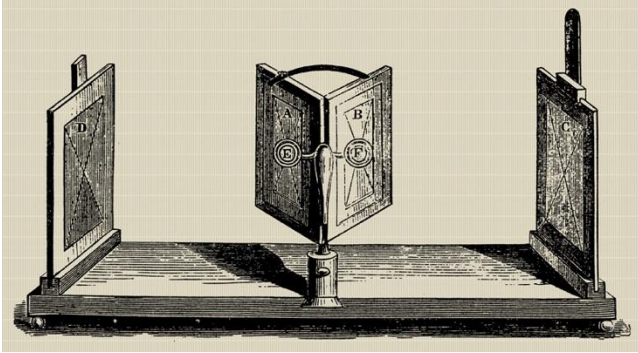
Human eyes **fixate** on point in space – rotate so that corresponding images form in centers of fovea.



[From Bruce and Green, Visual Perception, Physiology, Psychology and Ecology]

Stereo photography and stereo viewers

Take two pictures of the same subject from two slightly different viewpoints and display so that each eye sees only one of the images.



Invented by Sir Charles Wheatstone, 1838



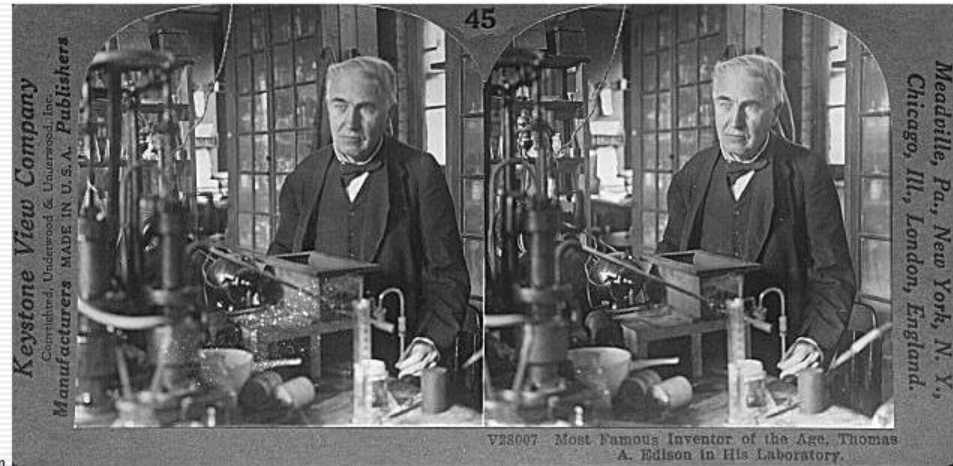
[Image from fisher-price.com]



Anaglyph Stereo

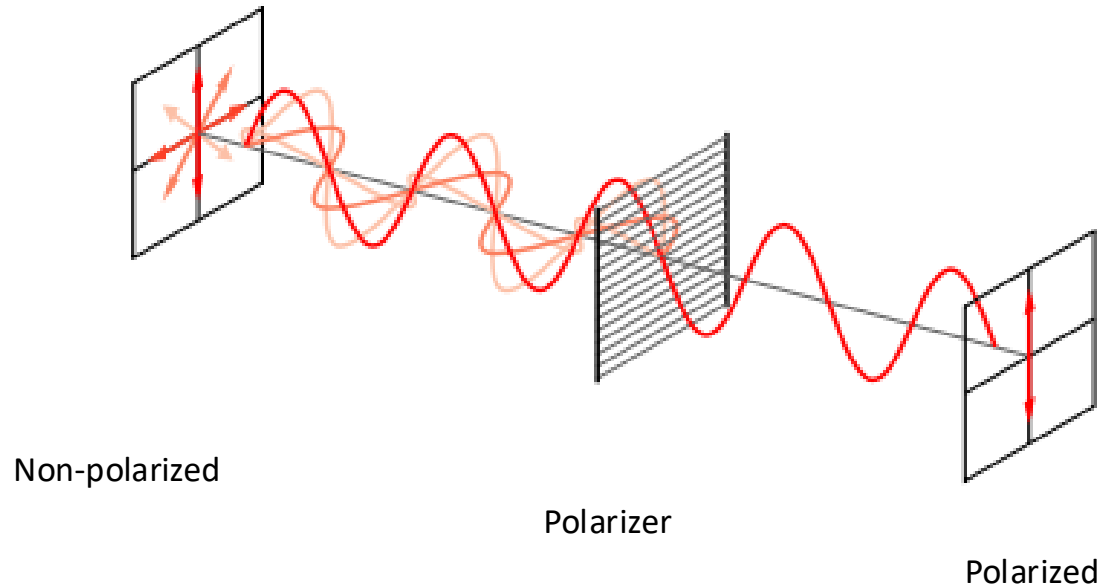


© Copyright 2001 Johnson-Shaw Stereoscopic Museum

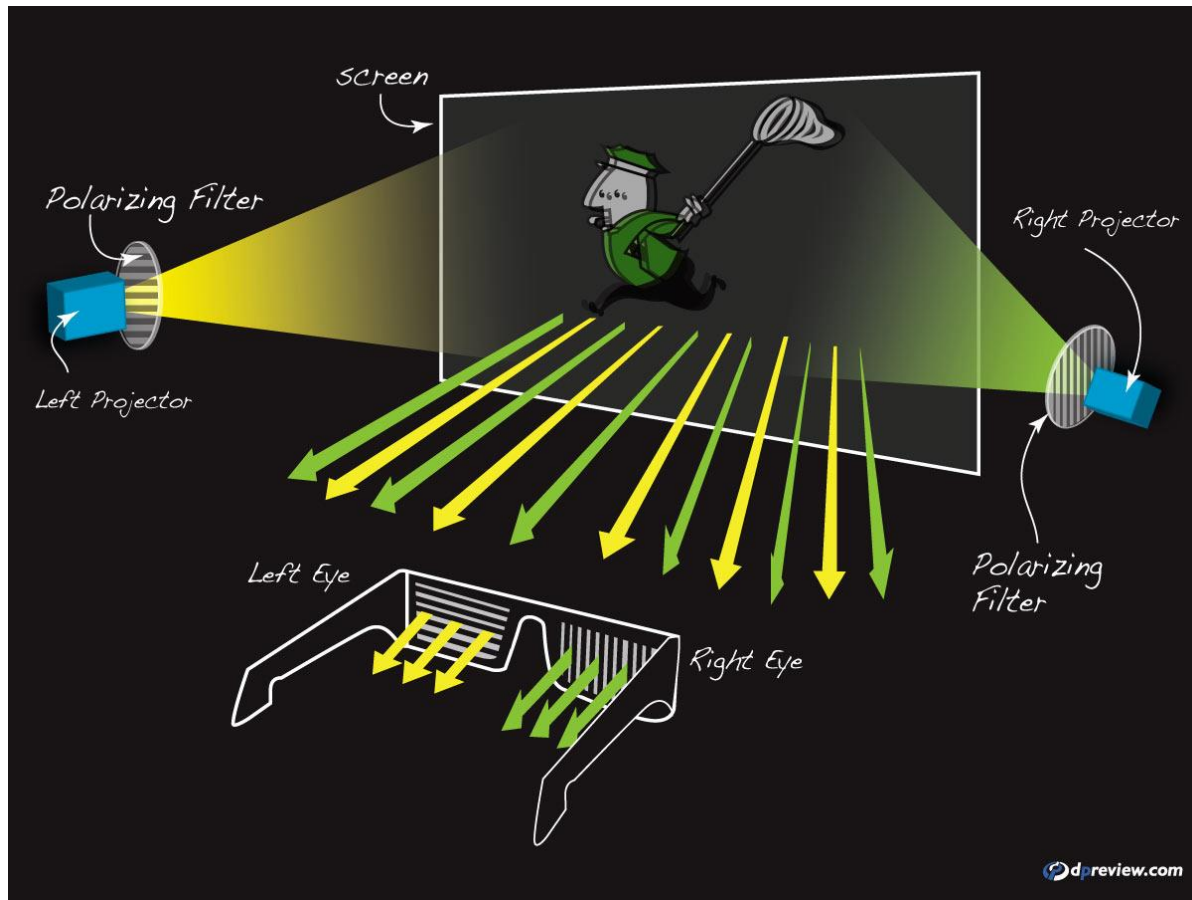


[<http://www.johnsonshawmuseum.org>]

3D Movie: View Separation via Polarization



3D Movie: View Separation via Polarization





Two cameras, simultaneous
views



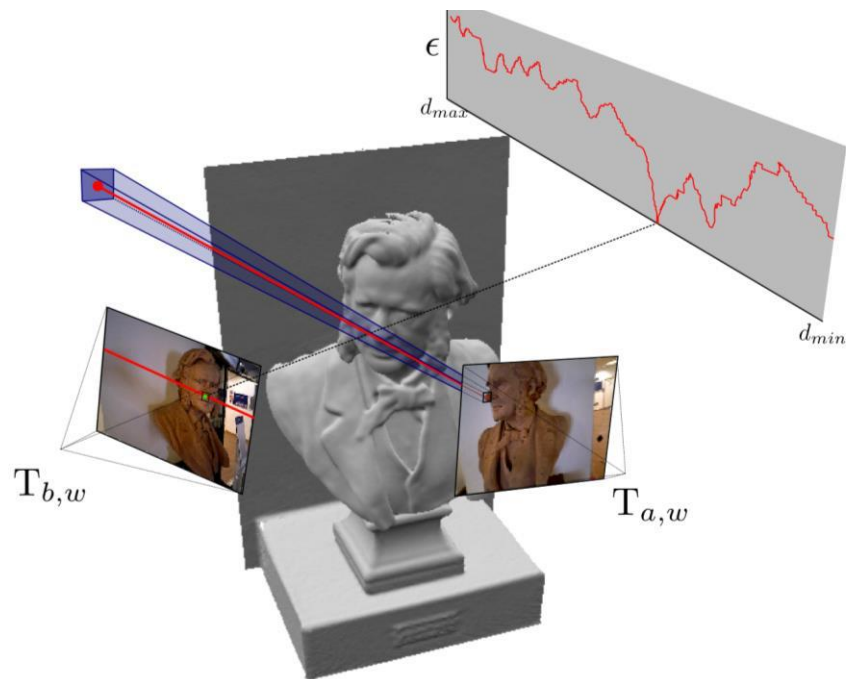
Single moving camera and
static scene

Why multiple views?

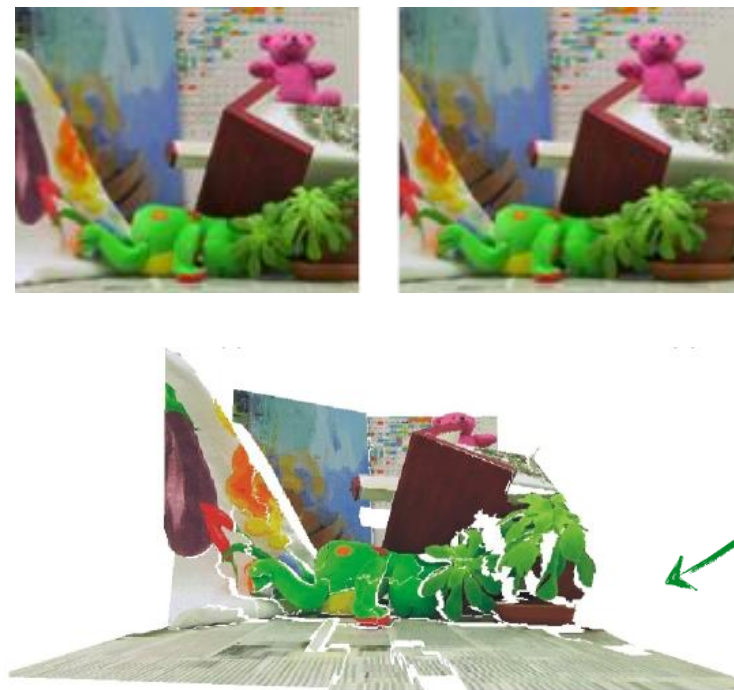
Structure and depth can be ambiguous from single views...



Multi-view Stereo

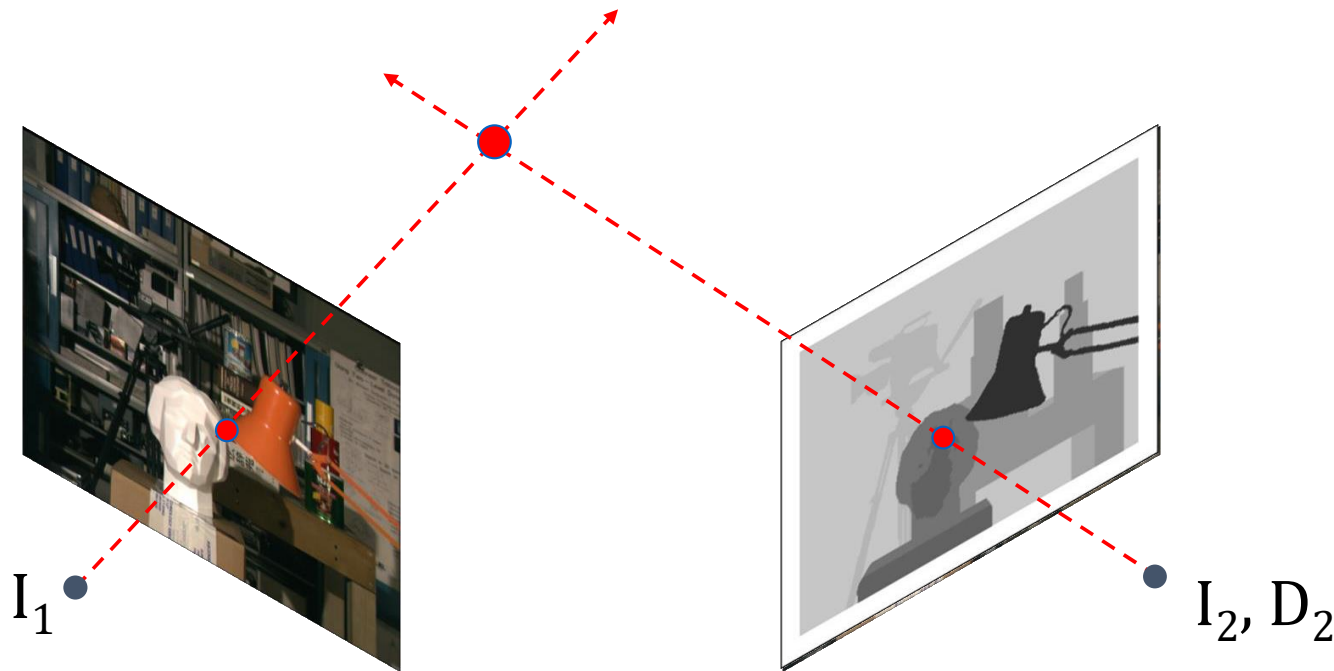


General Setting



Rectified Camera Setting

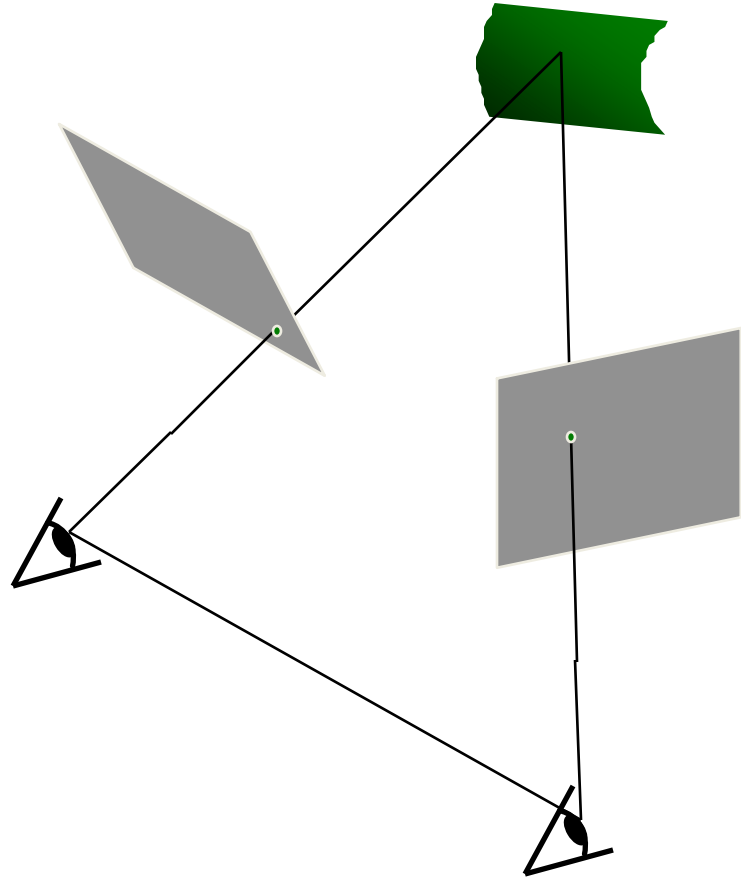
Dense Scene Geometry Generative Model



$$I_2(x,y) = I_1(\pi(T_{12} K^{-1} D_2(x,y) [x,y]))$$

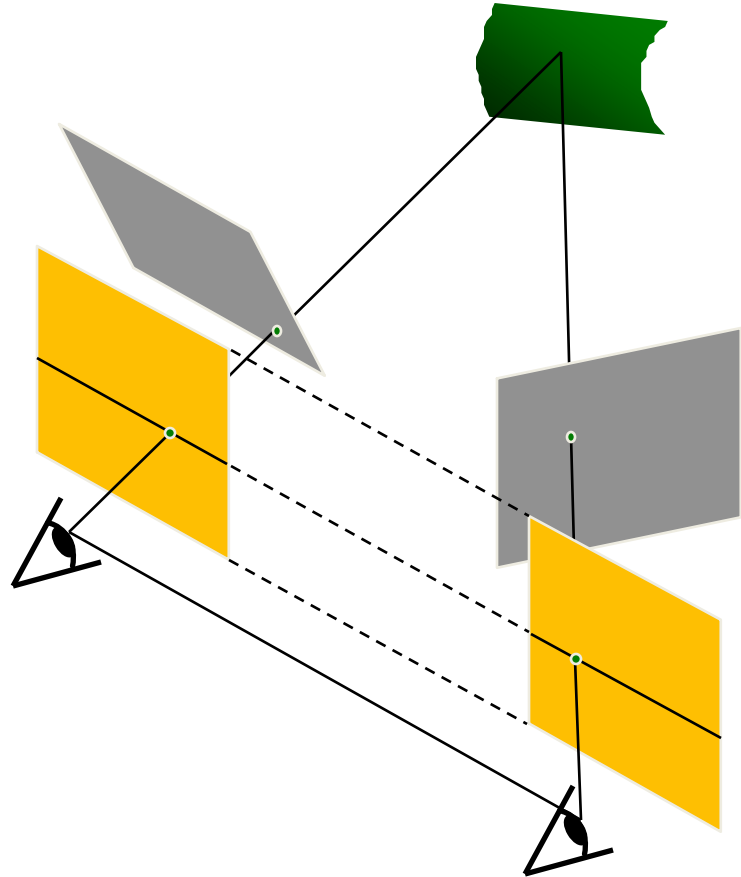
Stereo Image Rectification

- Project each image onto a common plane parallel to the baseline using homographies



Stereo Image Rectification

- Project each image onto a common plane parallel to the baseline using homographies



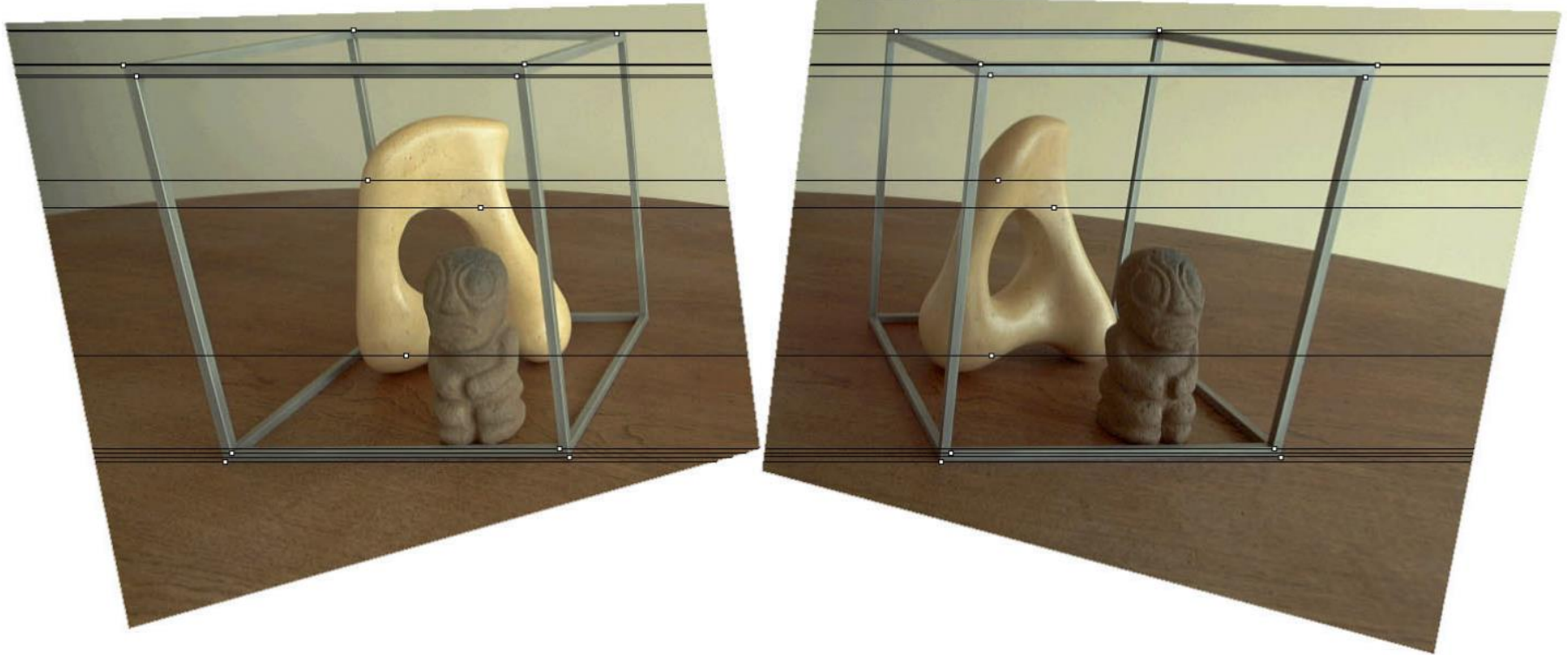
Stereo Image Rectification

Before rectification

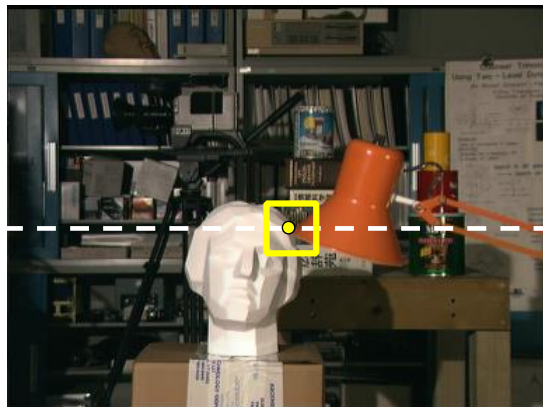
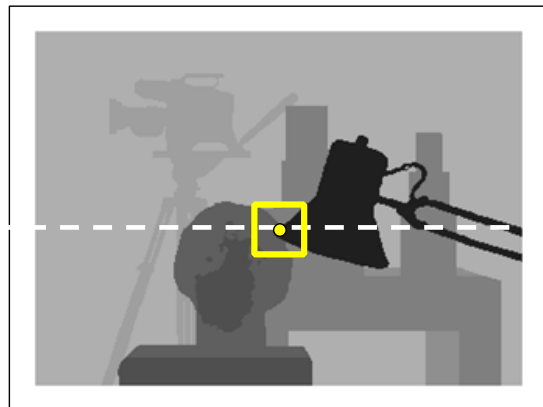
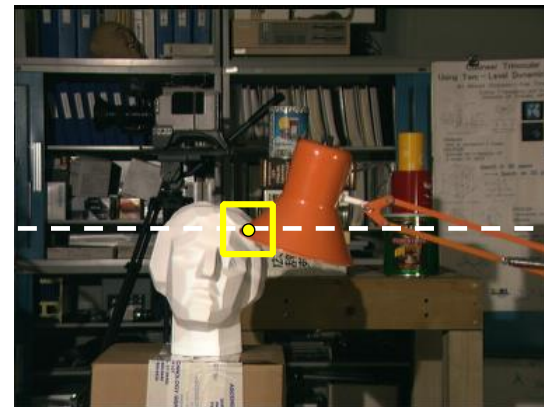


Stereo Image Rectification

After rectification



Special Case: Rectified Stereo Pair

 I_1 *Depth Image (D_2)* I_2

Brightness constancy assumption to find corresponding points:

$$I_2(x, y) = I_1 \left(x + \underbrace{d_2(x, y)}_{\text{Disparity}}, y \right)$$

Stereo Correspondence as Energy Minimization

 $I_1(x, y)$  $I_2(x, y)$

Pixel Error:

$$e(x, y, d) = I_1(x + d, y) - I_2(x, y)$$

Cost (with quadratic penalty):

$$C(x, y, d) = (I_1(x + d, y) - I_2(x, y))^2$$



Stereo Correspondence as Energy Minimization



Simple pixel / window matching: choose the minimum of each column in the DSI independently:

$$d(x, y) = \arg \min_{d'} C(x, y, d')$$

Stereo Correspondence as Energy Minimization

Effect of window size (W) for aggregating the photometric cost:

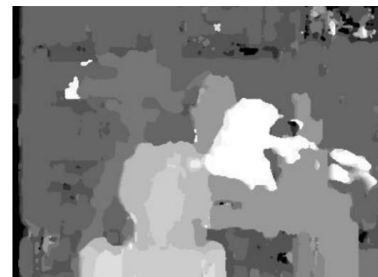
$$\sum_{(i,j) \in W} |I_1(i,j) - I_2(x+i, y+j)|$$



Ground truth



SAD W=3



SAD W=11



SAD W=25

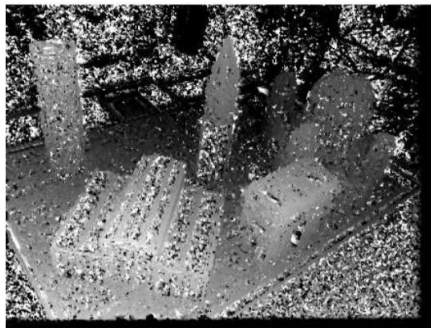
Stereo Correspondence as Energy Minimization

The design of the cost function, including *window size* for aggregation, *image error* function and *penalty* can improve quality of correspondence:

Similarity Measure	Formula
Sum of Absolute Differences (SAD)	$\sum_{(i,j) \in W} I_1(i,j) - I_2(x+i, y+j) $
Sum of Squared Differences (SSD)	$\sum_{(i,j) \in W} (I_1(i,j) - I_2(x+i, y+j))^2$
Zero-mean SAD	$\sum_{(i,j) \in W} I_1(i,j) - \bar{I}_1(i,j) - I_2(x+i, y+j) + \bar{I}_2(x+i, y+j) $
Locally scaled SAD	$\sum_{(i,j) \in W} I_1(i,j) - \frac{\bar{I}_1(i,j)}{\bar{I}_2(x+i, y+j)} I_2(x+i, y+j) $
Normalized Cross Correlation (NCC)	$\frac{\sum_{(i,j) \in W} I_1(i,j) \cdot I_2(x+i, y+j)}{\sqrt{\sum_{(i,j) \in W} I_1^2(i,j) \cdot \sum_{(i,j) \in W} I_2^2(x+i, y+j)}}$

Multi View Stereo

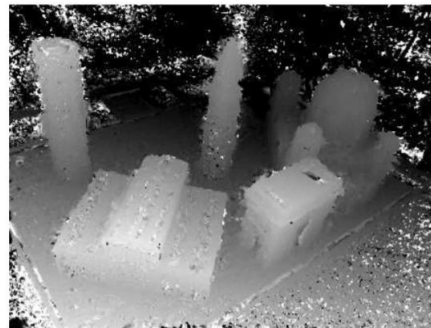
- How to Integrate more information from Multiple Views?



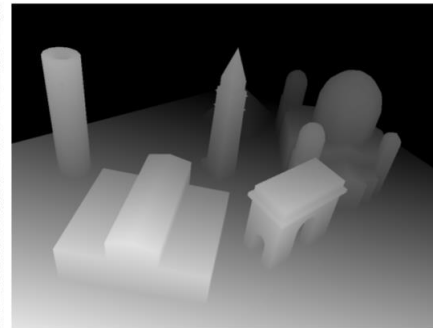
(a) 2 views



(b) 5 views



(c) 20 views



(d) Ground Truth

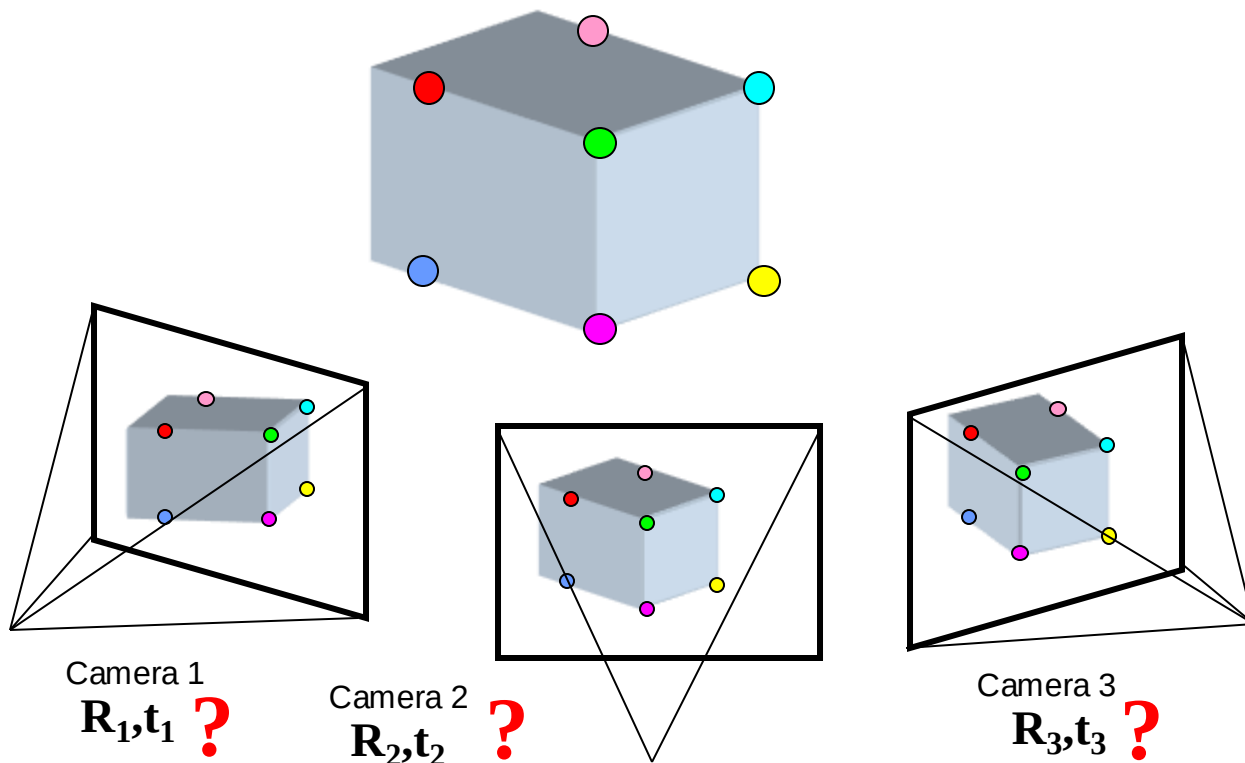
Today's Agenda



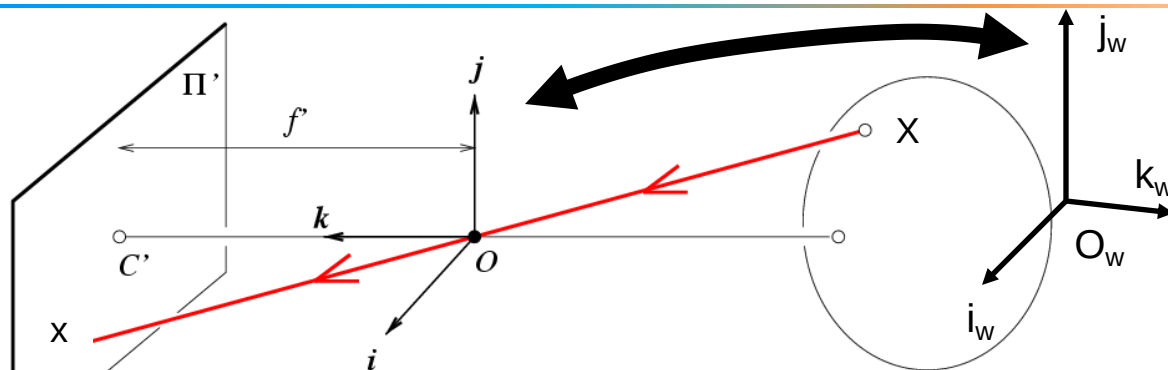
- Shape from X
- 3D Reconstruction
 - 2-view case: Stereo reconstruction
 - Camera calibration
 - Epipolar geometry
- Structure from Motion

Camera Calibration

Camera 'Motion': Given a set of corresponding 2D/3D points in two or more images, compute the camera parameters.



Recap: Camera Model



$$\mathbf{x} = \mathbf{K} \underbrace{\begin{bmatrix} \mathbf{R} & \mathbf{t} \end{bmatrix}}_{\text{Extrinsic Matrix}} \mathbf{X}$$

Extrinsic
Matrix

$\mathbf{x}$: Image Coordinates: $(u, v, 1)$

$\mathbf{K}$: Intrinsic Matrix (3x3)

$\mathbf{R}$: Rotation (3x3)

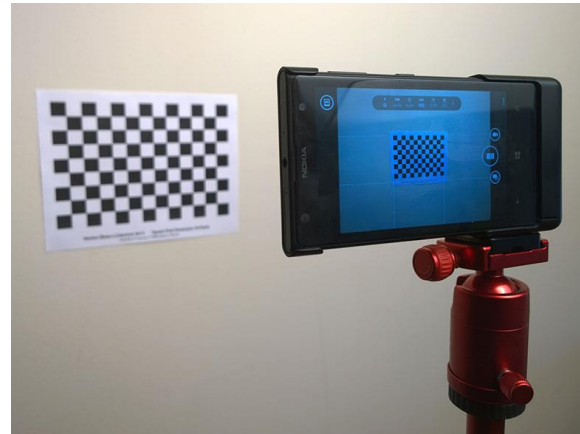
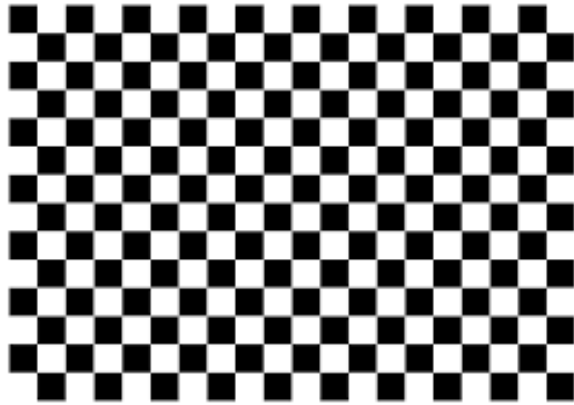
$\mathbf{t}$: Translation (3x1)

$\mathbf{X}$: World Coordinates: $(X, Y, Z, 1)$

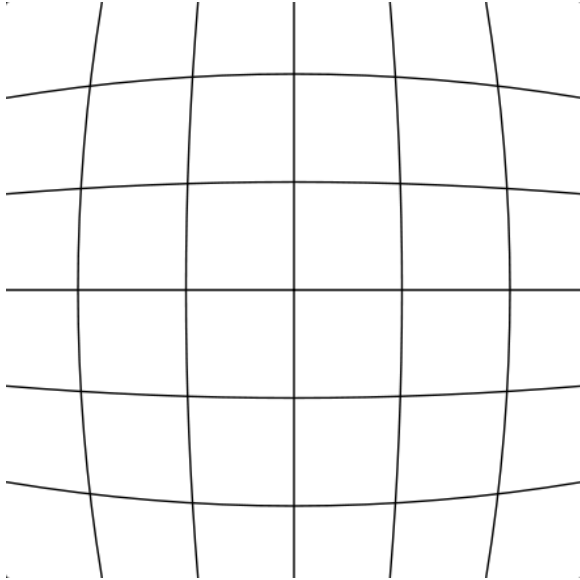
$${}^w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & s & u_0 \\ 0 & f_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \underbrace{\begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix}}_{\text{Extrinsic Matrix}} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

Intrinsic Calibration

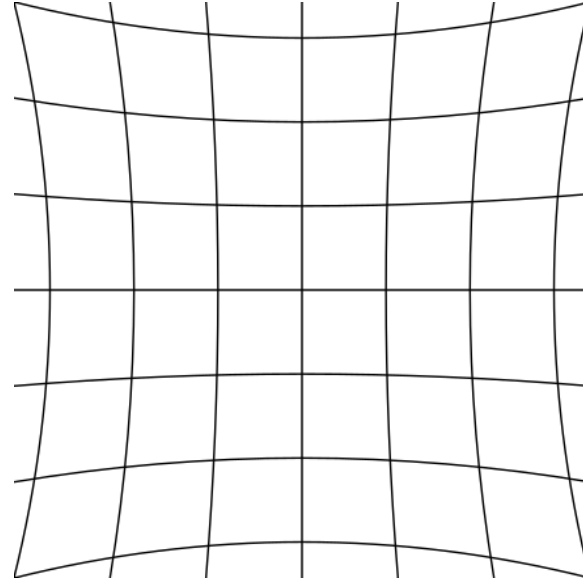
$$w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & s & u_0 \\ 0 & f_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$



Intrinsic Calibration: Non-linear Distortion

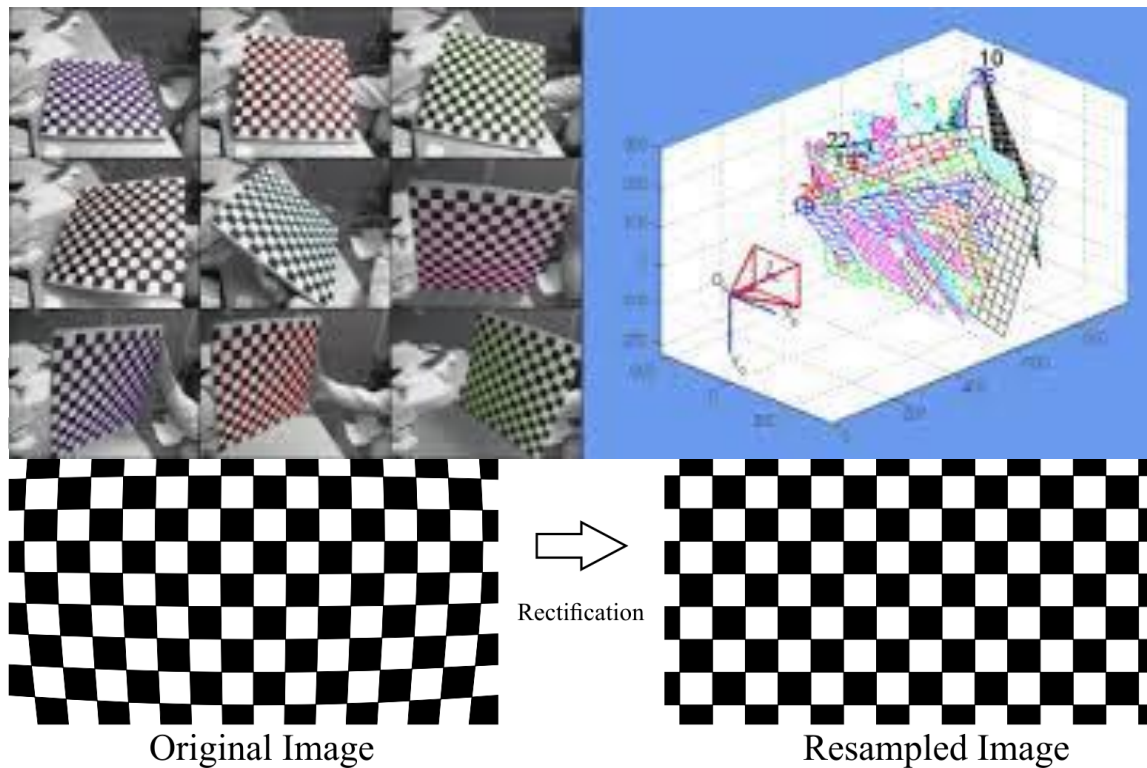


Barrel distortion



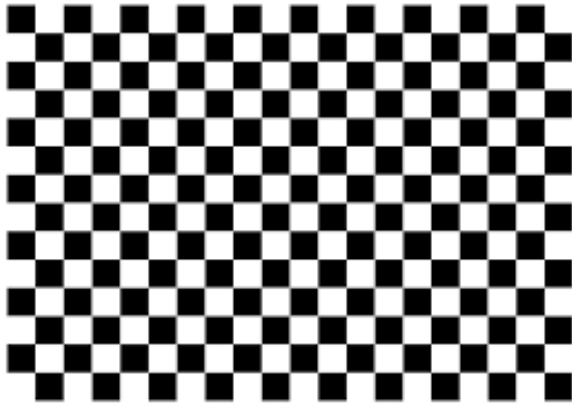
Pincushion distortion

Intrinsic Calibration: Non-linear Distortion



Extrinsic Calibration

$$w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & s & u_0 \\ 0 & f_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

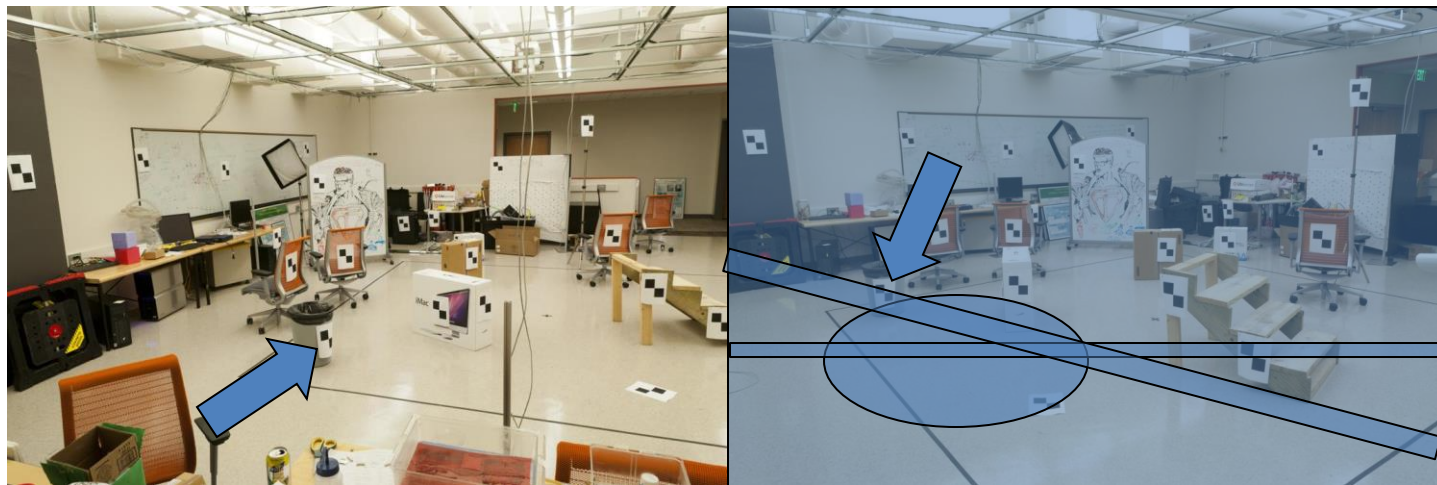


Today's Agenda



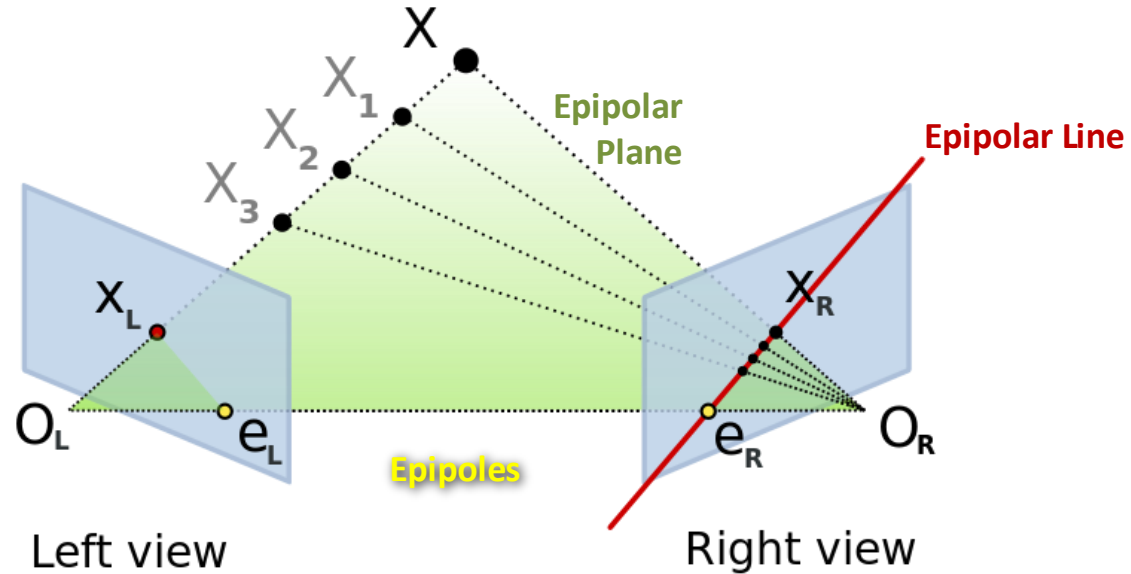
- Shape from X
- 3D Reconstruction
 - 2-view case: Stereo reconstruction
 - Camera calibration
 - Epipolar geometry
- Structure from Motion

Epipolar Geometry



Epipolar geometry constraints 2D search to 1D!

Epipolar Geometry

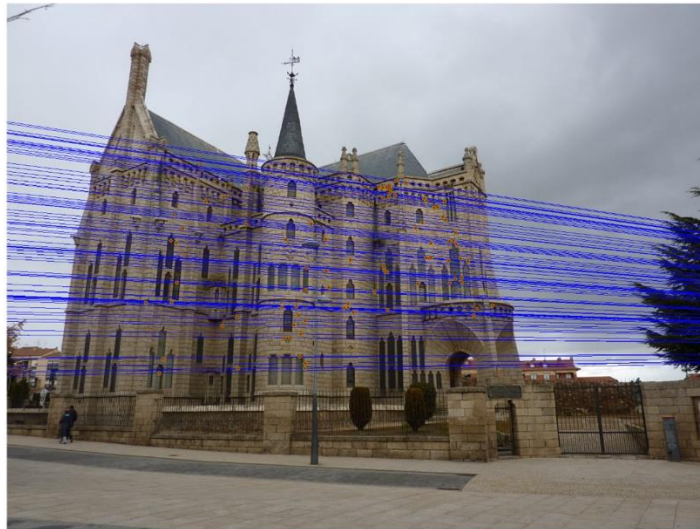
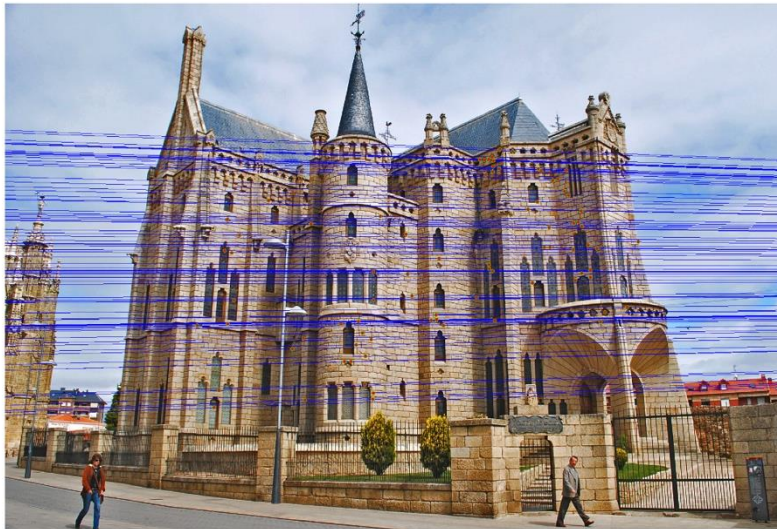


Epipolar Geometry

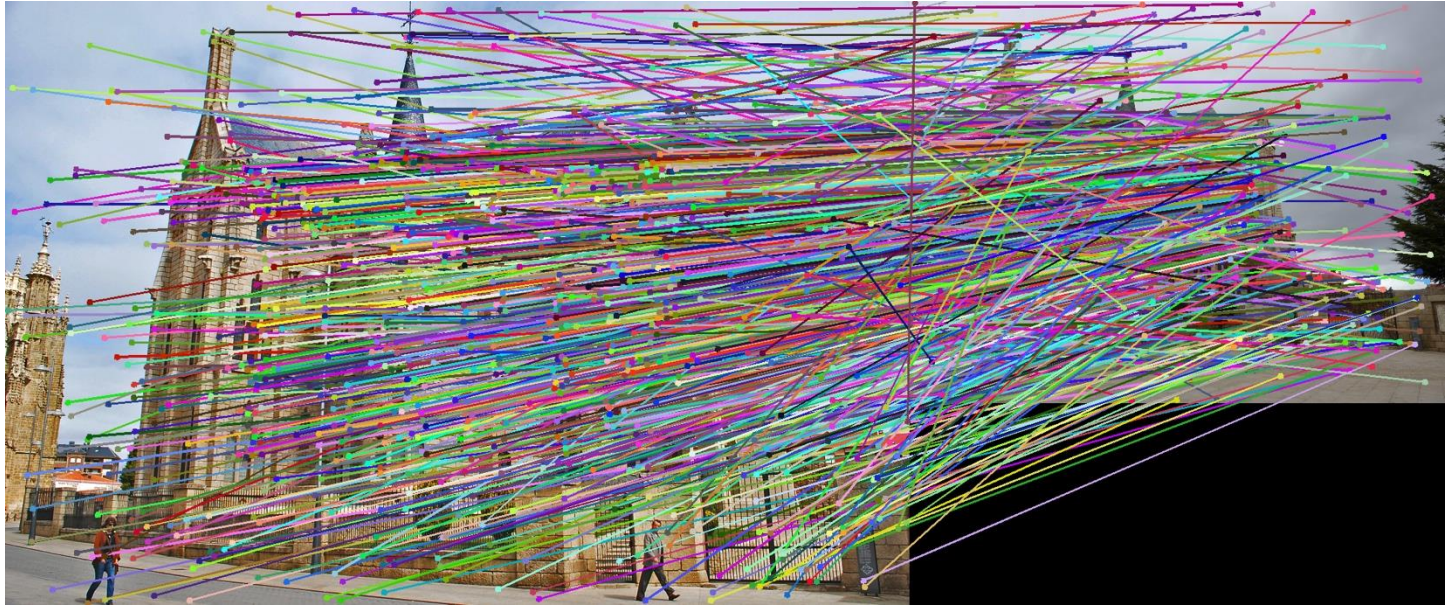


[Noah Snavely, Fundamental matrix demo, 2022.
<https://www.cs.cornell.edu/courses/cs5670/2022sp/demos/FundamentalMatrix/?demo=demo1>]

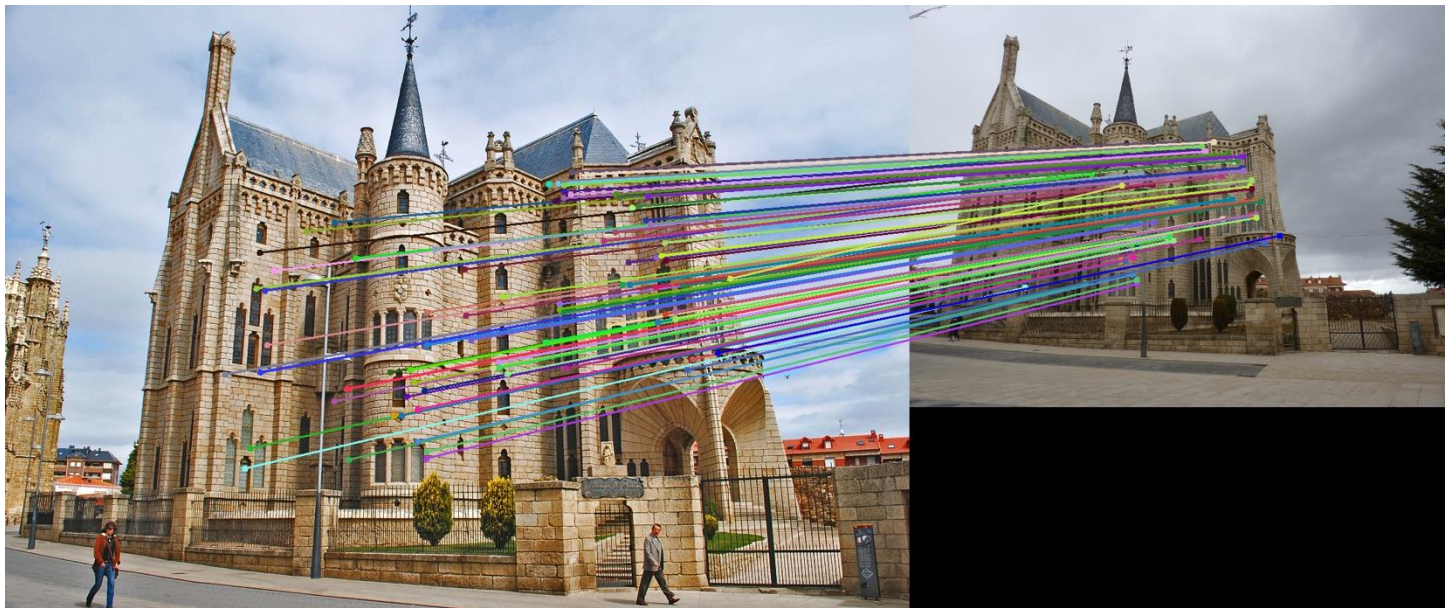
Epipolar Lines



Feature Correspondences



Feature Correspondences



Keep only the matches that are “inliers” with respect to the epipolar geometry (via RANSAC).

Today's Agenda

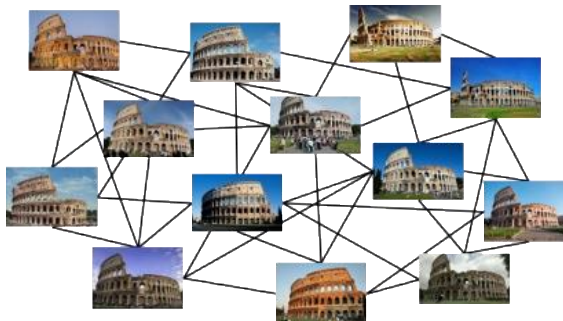
- Shape from X
- 3D Reconstruction
 - 2-view case: Stereo reconstruction
 - Camera calibration
 - Epipolar geometry
- Structure from Motion

Batch-based 3D Reconstruction Pipeline



Image Set

Image
Association



Scene Graph

SfM



Sparse Model

MVS



(Semi-) Dense Model

Structure-from-Motion (SfM)

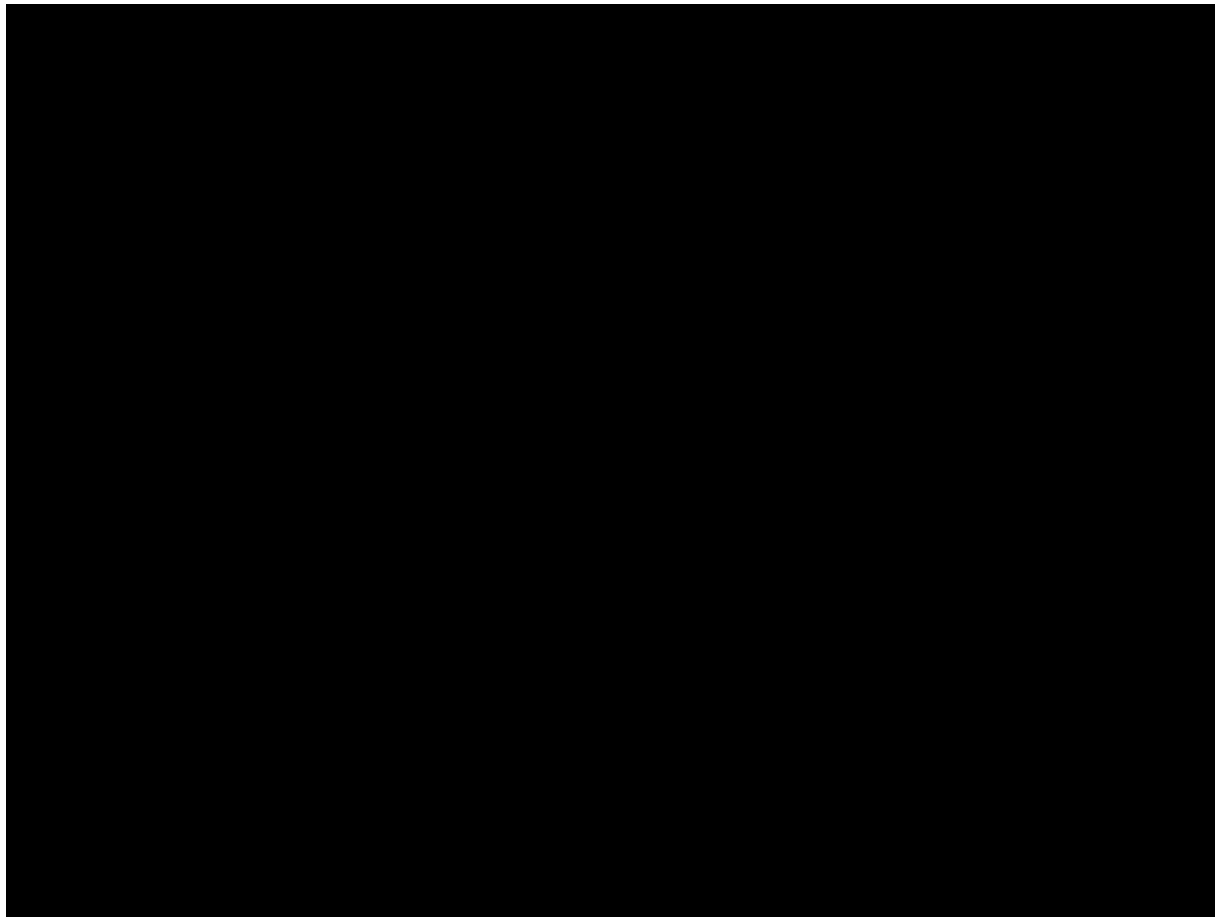
$$w \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & s & u_0 \\ 0 & f_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$$

- Mutli view geometry
- Structure from motion

Unknown

Unknown

Unknown



Rome dataset

74,394 images

Structure-from-Motion (SfM)

Available Structure from Motion Software

- Bundler <https://www.cs.cornell.edu/~snavey/bundler> by Noah Snavely (Open Source, Freeware)
- VisualSFM <http://ccwu.me/vsfm> by Changchang Wu (Closed Source, Freeware)
- COLMAP <https://colmap.github.io> by Johannes Schönberger (Open Source, Freeware)
- AliceVision <https://alicevision.org/> (Open Source, Freeware)
- Theia <http://www.theia-sfm.org> by Chris Sweeney (Open Source, Freeware)
- OpenMVG <https://github.com/openMVG/openMVG> by Pierre Moulon (Open Source, Freeware)

Disclaimer

Many of the slides used here are obtained from online resources (including many open lecture materials) without appropriate acknowledgement. They are used here for the sole purpose of classroom teaching. All the credit and all the copyrights belong to the original authors. You should not copy it, redistribute it, put it online, or use it for any other purposes than for this course.